微生物检测文献精读 · 非光学检测

等温微量热法无菌检测
从热流阈值到验证边界

BL0078 · 非光学检测：代谢热、报阳时间与对照设计

Indra Sioen · Tom Coenye

Evaluation of a Novel Isothermal Microcalorimetry-Based Sterility Test

在线发表 2026-08-24；卷期 2027 · 22:107

Sioen I, Coenye T. J Pharm Innov 22:107 (2027); online 2026-08-24. DOI: 10.1007/s12247-026-10986-0
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内容

检测原理与仪器

研究动机；热传导与热电转换；热流和累积热；阈值与报阳规则。

数据与统计

全部主文图表、补充图表；报阳比例、时间分布、漏检与精密度。

输入与控制

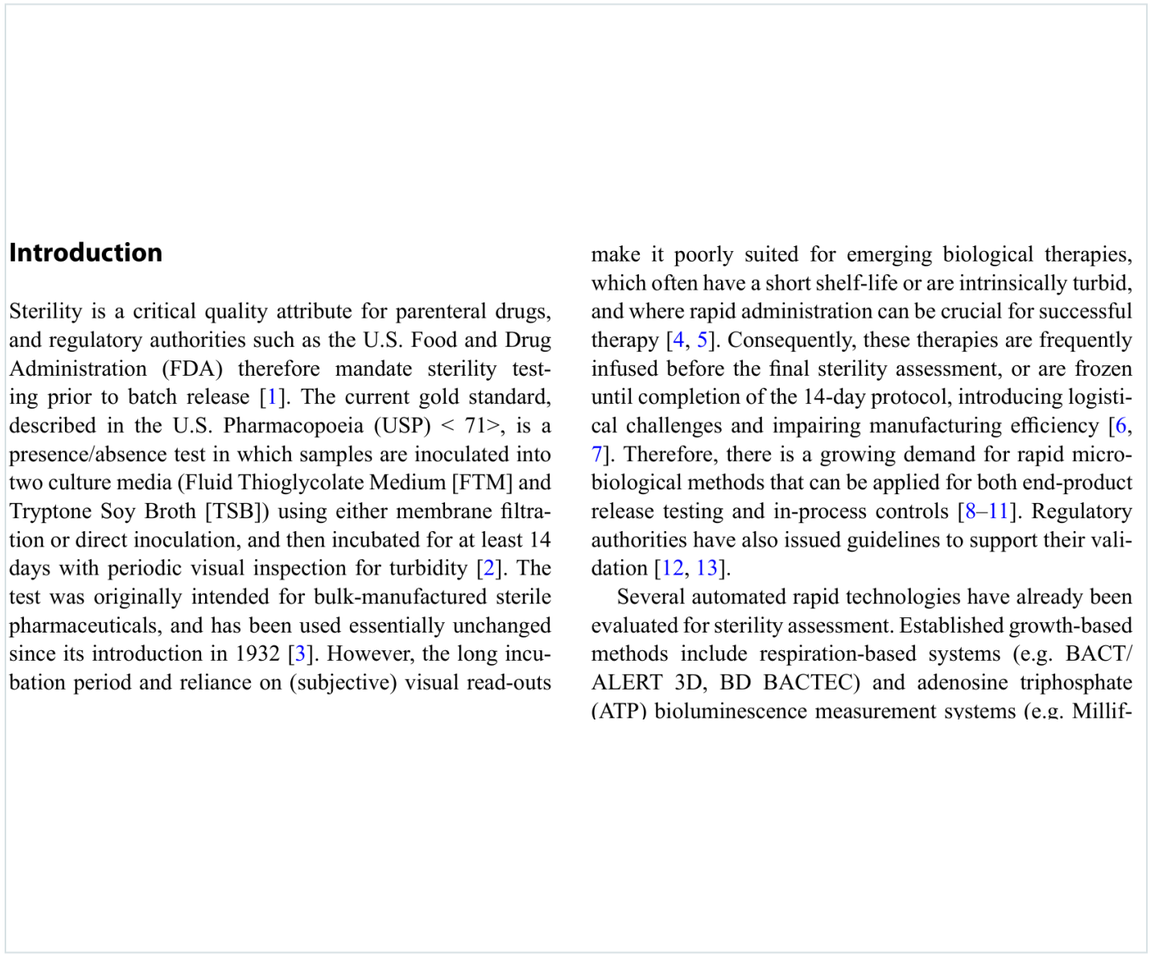
16种微生物、实际挑战量、空白基线、技术重复与生物学重复。

论文储备

可用于introduction的证据；设计混杂、未验证指标与设备研究假设。

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检测任务：更早发现污染，同时保留阴性判定边界



Introduction

Sterility is a critical quality attribute for parenteral drugs, and regulatory authorities such as the U.S. Food and Drug Administration (FDA) therefore mandate sterility testing prior to batch release [1]. The current gold standard, described in the U.S. Pharmacopoeia (USP) < 71>, is a presence/absence test in which samples are inoculated into two culture media (Fluid Thioglycolate Medium [FTM] and Tryptone Soy Broth [TSB]) using either membrane filtration or direct inoculation, and then incubated for at least 14 days with periodic visual inspection for turbidity [2]. The test was originally intended for bulk-manufactured sterile pharmaceuticals, and has been used essentially unchanged since its introduction in 1932 [3]. However, the long incubation period and reliance on (subjective) visual read-outs make it poorly suited for emerging biological therapies, which often have a short shelf-life or are intrinsically turbid, and where rapid administration can be crucial for successful therapy [4, 5]. Consequently, these therapies are frequently infused before the final sterility assessment, or are frozen until completion of the 14-day protocol, introducing logistical challenges and impairing manufacturing efficiency [6, 7]. Therefore, there is a growing demand for rapid microbiological methods that can be applied for both end-product release testing and in-process controls [8–11]. Regulatory authorities have also issued guidelines to support their validation [12, 13].

Several automated rapid technologies have already been evaluated for sterility assessment. Established growth-based methods include respiration-based systems (e.g. BACT/ALERT 3D, BD BACTEC) and adenosine triphosphate (ATP) bioluminescence measurement systems (e.g. Millif-

原文截图 · PDF p.1, Introduction

论文面对的需求

作者讨论短货架期制品与内在浑浊样品：依赖目视浊度的检测既有等待时间，也有读出主观性。

本研究比较什么

calScreener+ 连续记录代谢热；参照方法每日目视检查。研究是16种微生物的概念验证，不是实际药品放行试验。

更早报阳可以提前提示污染；平均报阳更快，不能直接规定所有阴性样品的缩短放行时限。

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非光学信号：代谢活动产生热功率

Heat is a byproduct of the metabolism of microorganisms, and isothermal microcalorimetry (IMC) devices allow continuous measurement of this heat production in real-time. As IMC measures a fundamental physiological parameter, the method is inherently label-free and suitable for diverse sample types, including turbid, viscous or solid samples. In addition, IMC is a non-destructive method, allowing downstream analysis of the sample. Moreover, several studies have shown that IMC can detect microorganisms earlier than other growth-based methods while maintaining a high sensitivity [19, 20]. In the present study, we conducted a proof-of-concept evaluation of the calScreener + IMC device (Symcel) for sterility testing and compared its performance against the USP < 71 > compendial sterility test in terms of number of positive cultures detected and TTD.

原文截图 · PDF p.2, Introduction, metabolic heat

输入、输出与仪器接口

输入是密闭测量容器中的样品体系。代谢过程释放热，仪器输出热流随时间的曲线：纵轴 μW，横轴 h。

测量对象

热流是能量释放速率。它无需荧光或透射光，但并非微生物特异的化学指纹；其他放热过程也可能构成背景。

论文将热流超过1 μW定义为阳性。本文没有用热曲线识别菌种，也没有直接输出活菌数。

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热传导与热电转换的物理映射

maximal baseline stability). Metabolic heat production was monitored in real-time, generating thermograms that plot heat flow (μW) over time (calView2 software, Symcel). Samples were considered positive if the heat flow exceeded 1 μW. The timepoint at which this occurred was

原文截图 · PDF p.2, Methods, thermal readout and threshold (text excerpt)

教学示意，非原文图

样品放热 → 热传导路径 → 热电堆电压 → 参照/基线处理 → 校准热流 P(t)。

等温指环境近恒温；微小温差仍驱动传热并被读出。

$P \approx G\Delta T$；$U - U_0 = S\Delta T$

$P \approx (G/S)(U - U_0) = k(U - U_0)$

符号、有效条件与缺失参数

P：热功率（W）；G：热导（W/K）；ΔT：微小温差（K）；S：热电灵敏度（V/K）；U、U_0：信号与基线电压（V）；k：校准系数（W/V）。

近稳态、稳定热路径与有效校准下成立。热惯性会延迟快速变化。本文未公开G、S、k或响应时间，不能据此还原具体传感器。

热电堆与热沉背景：N. Vendur，Evaluation of an alternative microbiological technique based on microcalorimetry，2025，厂商公开会议资料 pp.7–8；仅引用原理，未复制其图片。

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演算：热功率、累积热和细胞数不可混用

maximal baseline stability). Metabolic heat production was monitored in real-time, generating thermograms that plot heat flow (μW) over time (calView2 software, Symcel). Samples were considered positive if the heat flow exceeded 1 μW. The timepoint at which this occurred was

原文截图 · PDF p.2, Methods, thermal readout and threshold (text excerpt)

假设教学数值，非论文数据

假设 k=0.50 W/V，$U-U_0$=2.4 μV，则：

$P=0.50 \times 2.4\times10^{-6}$ W
=1.2 μW

若该功率恒定维持10 min：

$Q=\int Pdt=1.2\ \mu J/s \times 600\ s$
=720 μJ=0.72 mJ

输出的解释

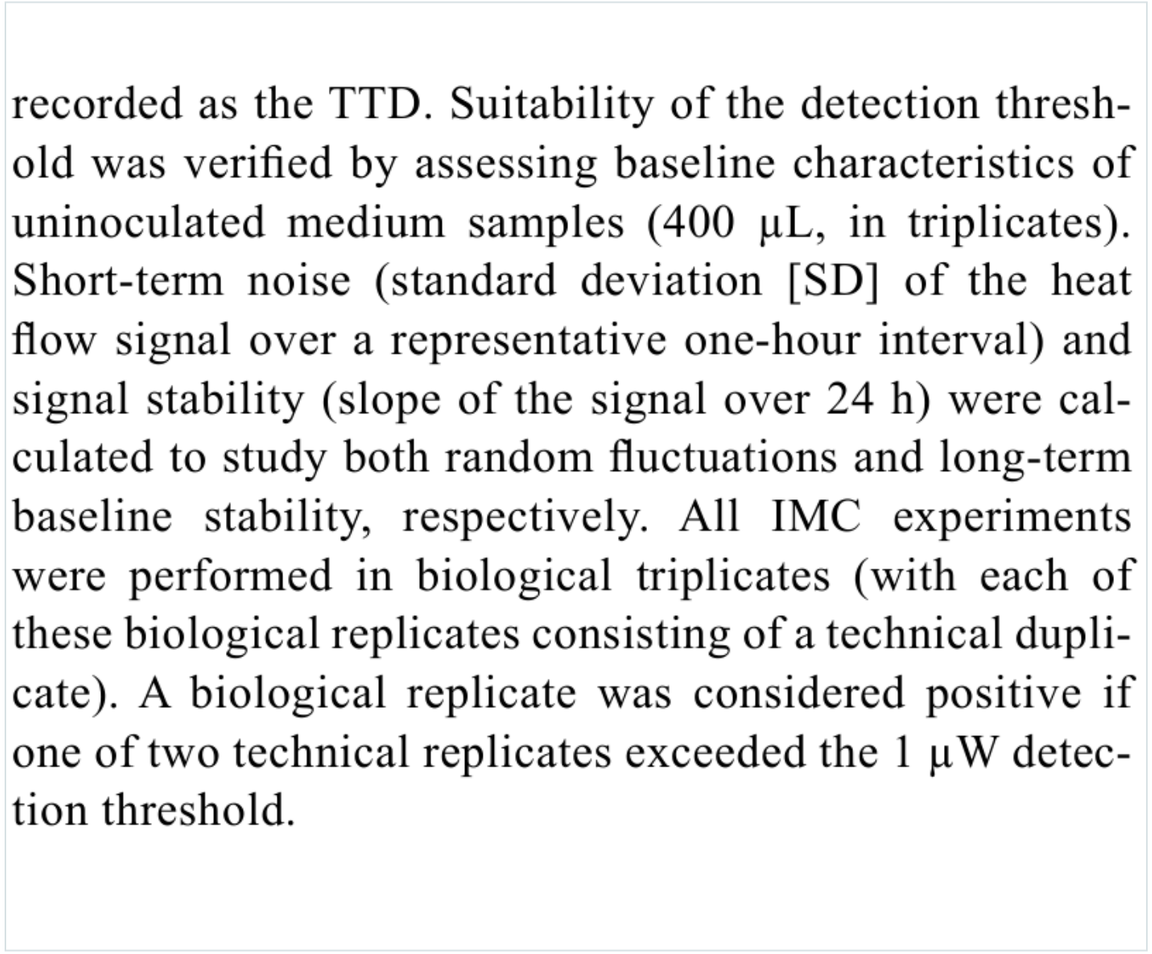
P是热功率；Q是累积热量（J）；t为时间（s）。1 W=1 J/s。

1.2 μW已超过本文1 μW阈值。0.72 mJ只说明积分能量，不能在未知单细胞产热率时换算为CFU。

真实曲线并非恒定，积分需使用采样时间；本文判阳使用热流阈值，不使用累积热阈值。

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报阳算法与计数单位



recorded as the TTD. Suitability of the detection threshold was verified by assessing baseline characteristics of uninoculated medium samples (400 µL, in triplicates). Short-term noise (standard deviation [SD] of the heat flow signal over a representative one-hour interval) and signal stability (slope of the signal over 24 h) were calculated to study both random fluctuations and long-term baseline stability, respectively. All IMC experiments were performed in biological triplicates (with each of these biological replicates consisting of a technical duplicate). A biological replicate was considered positive if one of two technical replicates exceeded the 1 µW detection threshold.

原文截图 · PDF p.3, Methods, baseline and positivity rule

论文的判定规则

TTD = 首次满足 P(t) > 1 µW 的时间

TTD 为 time to detection。每个生物学重复包含两个IMC技术重复；任一个技术重复超过阈值，该生物学重复就判阳。

假设教学案例

同一生物学重复的两条曲线分别在18 h和20 h过阈值：两支均有TTD，但生物学阳性计数只增加1。

论文没有明确描述两支技术TTD合并为该生物学重复TTD的全部细节。不要擅自规定“取最早”或“取平均”。

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验证设计与样本分母

In terms of experimental setup, every biological replicate in the microcalorimetric method consisted of two technical replicates, in line with standard practice and manufacturer recommendations. In contrast, the compendial method was performed using a single technical replicate. Technical duplicates are typically not included in compendial sterility testing. While our setup thus reflects routine practice for each method, it could partially explain the detection differences observed. Furthermore, our approach was designed to have the same number of CFU for both methods. As different volumes are used in both methods (5 mL for compendial vs. 400 µL for IMC), the CFU concentration is lower in the compendial method, potentially affecting its performance. On the other hand, as the maximum volume of the calScreener+ method is currently limited to 400 µL, this might in some cases complicate compliance with USP sample volume requirements. Possible approaches to meet the required volume could include dividing a sample over

原文截图 · PDF p.7, Discussion, technical replication and volumes

比较结构

16种微生物 × 2个名义挑战水平 × 3个生物学重复＝每方法96个生物学结果。每个水平的分母为48。

IMC每个生物学重复有2支技术重复，共192支；参照方法每个生物学重复为1支。两方法从同一微生物悬液并行取样。

不能合并的单位

技术重复不是新增独立生物学样本。研究没有训练/测试划分，也没有训练机器学习模型。

培养条件在原表中完整保留作验证背景；本课件不提供生物培养操作建议。

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Table 1：全部16种微生物与来源

Table 1 List of microorganisms used, with culture conditions

Species	Strain designation	Additional information	Media and culture conditions
Acidovorax delafieldii	LPM-S1	Isolated from pharmaceutical production environment (Belgium)	Tryptone Soy Agar (TSA)[a]; Nutrient Broth (NB)[a] Aerobic[b]
Aspergillus brasiliensis	ATCC 16404	USP<71>reference strain	Sabouraud Dextrose Agar (SDA)[a] Aerobic[b]
Aspergillus versicolor	LPM-S2	Isolated from pharmaceutical production environment (Belgium)	SDA Aerobic[b]
Bacillus spizizenii	ATCC 6633	USP<71>reference strain	TSA; NB Aerobic[b]
Bacillus subtilis	LPM-S3	Isolated from pharmaceutical production environment (Belgium)	TSA; NB Aerobic[b]
Burkholderia contaminans	LPM-S4	Isolated from contaminated liquid docusate	TSA; NB Aerobic[b]
Candida albicans	ATCC 10231	USP<71>reference strain	SDA, Sabouraud Dextrose Broth[c] Aerobic[b]
Clostridium sporogenes	ATCC 19404	USP<71>reference strain	Columbia Agar supplemented with 5% sheep blood (CBA)[a]; Reinforced Clostridial Medium (RCM)[c] Anaerobic[b]
Cutibacterium acnes	LMG 16711	Isolated from human skin (patient with facial acne)	CBA; RCM Anaerobic[d]
Micrococcus luteus	LPM-S5	Isolated from human saliva	CBA; NB Aerobic[b]
Penicillium chrysogenum	LPM-S6	Isolated from pharmaceutical production environment (Belgium)	SDA Aerobic[b]
Pseudomonas paraeruginosa	ATCC 9027	USP<71>reference strain	TSA; NB Aerobic[b]
Staphylococcus aureus	ATCC 6538	USP<71>reference strain	TSA; NB Aerobic[b]
Staphylococcus epidermidis	LPM-S7	Isolated from pharmaceutical production environment (Belgium)	TSA; NB Aerobic[b]
Staphylococcus hominis	LPM-S8	Isolated from pharmaceutical production environment (Belgium)	TSA; NB Aerobic[b]
Streptococcus pyogenes	LMG 14237	Isolated from throat of a patient suffering from rheumatic fever	TSA; NB Aerobic[b]

[a] Supplier: Neogen
[b] Overnight culture: 37 °C, 16 h
[c] Supplier: Lab M
[d] Overnight culture: 37 °C, 40 h

原文截图 · PDF p.3, Table 1 complete

完整表定位：6株药典参考株，10株其他分离株；构成为9种革兰阳性菌、3种革兰阴性菌、3种霉菌及1种酵母。具体行与脚注在后两页放大。

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Table 1上半：参考株与环境/产品分离株

Species	Strain designation	Additional information	Media and culture conditions
Acidovorax delafieldii	LPM-S1	Isolated from pharmaceutical production environment (Belgium)	Tryptone Soy Agar (TSA)[a]; Nutrient Broth (NB)[a] Aerobic[b]
Aspergillus brasiliensis	ATCC 16404	USP<71>reference strain	Sabouraud Dextrose Agar (SDA)[a] Aerobic[b]
Aspergillus versicolor	LPM-S2	Isolated from pharmaceutical production environment (Belgium)	SDA Aerobic[b]
Bacillus spizizenii	ATCC 6633	USP<71>reference strain	TSA; NB Aerobic[b]
Bacillus subtilis	LPM-S3	Isolated from pharmaceutical production environment (Belgium)	TSA; NB Aerobic[b]
Burkholderia contaminans	LPM-S4	Isolated from contaminated liquid docusate	TSA; NB Aerobic[b]
Candida albicans	ATCC 10231	USP<71>reference strain	SDA, Sabouraud Dextrose Broth[c] Aerobic[b]
Clostridium sporogenes	ATCC 19404	USP<71>reference strain	Columbia Agar supplemented with 5% sheep blood (CBA)[a]; Reinforced Clostridial Medium (RCM)[c] Anaerobic[b]

原文截图 · PDF p.3, Table 1 rows 1–8 and column headings

前8项覆盖 A. delafieldii、两种Aspergillus、两种Bacillus、B. contaminans、C. albicans及C. sporogenes。右侧培养条件是研究输入背景，不是传感器性能参数。

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Table 1下半：来源多样不等于真实基质验证

Cutibacterium acnes	LMG 16711	Isolated from human skin (patient with facial acne)	CBA; RCM Anaerobic[d]
Micrococcus luteus	LPM-S5	Isolated from human saliva	CBA; NB Aerobic[b]
Penicillium chrysogenum	LPM-S6	Isolated from pharmaceutical production environment (Belgium)	SDA Aerobic[b]
Pseudomonas paraeruginosa	ATCC 9027	USP<71>reference strain	TSA; NB Aerobic[b]
Staphylococcus aureus	ATCC 6538	USP<71>reference strain	TSA; NB Aerobic[b]
Staphylococcus epidermidis	LPM-S7	Isolated from pharmaceutical production environment (Belgium)	TSA; NB Aerobic[b]
Staphylococcus hominis	LPM-S8	Isolated from pharmaceutical production environment (Belgium)	TSA; NB Aerobic[b]
Streptococcus pyogenes	LMG 14237	Isolated from throat of a patient suffering from rheumatic fever	TSA; NB Aerobic[b]

[a] Supplier: Neogen
[b] Overnight culture: 37 °C, 16 h
[c] Supplier: Lab M
[d] Overnight culture: 37 °C, 40 h

原文截图 · PDF p.3, Table 1 rows 9–16 and footnotes

后8项包含皮肤、唾液、咽部与制药环境来源；所有来源、菌株号及a–d脚注保留。来源于临床或产品的分离株，不能替代直接在临床样品或成品制剂中的验证。

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Online Resource 1：实际挑战量完整表

Online Resource 1 - Table Overview of all tested organisms with organism type and plate counts for 50 µL of suspension (mean ± standard deviation; n = 6).

		Plate counts (CFU)	
Organism	**Organism type**	**Aim 100 CFU**	**Aim 5 CFU**
***Acidovorax delafieldii* LPM-S1**	Gram-negative	30 ± 15	3 ± 2
***Aspergillus brasiliensis* ATCC 16404**	Mold	45 ± 22	3 ± 1
***Aspergillus versicolor* LPM-S2**	Mold	39 ± 10	3 ± 2
***Bacillus spizizenii* ATCC 6633**	Gram-positive	40 ± 12	2 ± 1
***Bacillus subtilis* LPM-S3**	Gram-positive	65 ± 18	2 ± 1
***Burkholderia contaminans* LPM-S4**	Gram-negative	65 ± 6	3 ± 2
***Candida albicans* ATCC 10231**	Yeast	52 ± 16	4 ± 2
***Clostridium sporogenes* ATCC 19404**	Gram-positive	37 ± 17	3 ± 2
***Cutibacterium acnes* LMG 16711**	Gram-positive	181 ± 49	11 ± 6
***Micrococcus luteus* LPM-S5**	Gram-positive	120 ± 24	7 ± 1
***Penicillium chrysogenum* LPM-S6**	Mold	55 ± 5	4 ± 2
***Pseudomonas paraeruginosa* ATCC 9027**	Gram-negative	72 ± 38	4 ± 2
***Staphylococcus aureus* ATCC 6538**	Gram-positive	89 ± 17	6 ± 3
***Staphylococcus epidermidis* LPM-S7**	Gram-positive	39 ± 15	3 ± 2
***Staphylococcus hominis* LPM-S8**	Gram-positive	36 ± 5	3 ± 2
***Streptococcus pyogenes* LMG 14237**	Gram-positive	22 ± 9	2 ± 1

Abbreviations: CFU, colony-forming unit

原文截图 · Supplement PDF p.1, Online Resource 1, complete table

表中每个均值来自6个平板计数（3个生物学重复的计数技术重复），误差为SD。名义“100 CFU”和“5 CFU”是组名；各物种的实际计数并不相同。

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实际输入的偏离与比较范围

***Bacillus spizizenii* ATCC 6633**	Gram-positive	40 ± 12	2 ± 1
***Bacillus subtilis* LPM-S3**	Gram-positive	65 ± 18	2 ± 1
***Burkholderia contaminans* LPM-S4**	Gram-negative	65 ± 6	3 ± 2
***Candida albicans* ATCC 10231**	Yeast	52 ± 16	4 ± 2
***Clostridium sporogenes* ATCC 19404**	Gram-positive	37 ± 17	3 ± 2
***Cutibacterium acnes* LMG 16711**	Gram-positive	181 ± 49	11 ± 6

原文截图 · Supplement PDF p.1, Online Resource 1 rows 4–9; numeric columns: aim 100 CFU, aim 5 CFU

原文报告与全表范围

高水平实际单次计数范围22–250 CFU，总体62±47；低水平1–19 CFU，总体4±3。原文的组名不能当成每次真实输入。

补表物种均值中，C. acnes为181±49及11±6；两种Bacillus低水平均为2±1。

对指标的影响

用同一悬液并行比较减少输入差异，但不同物种的TTD并非在完全相同CFU下比较。

补表的n=6是平板计数重复，不是每方法有6个独立生物学重复。现有两档挑战量不足以建立正式LoD。

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热力学参照与未接种空白的不同作用

suspensions. Thermodynamic reference vials were filled with 400 µL of PS. Vials were sealed and incubated in

原文截图 · PDF p.2, Methods, thermodynamic reference (text excerpt)

recorded as the TTD. Suitability of the detection threshold was verified by assessing baseline characteristics of uninoculated medium samples (400 µL, in triplicates). Short-term noise (standard deviation [SD] of the heat flow signal over a representative one-hour interval) and signal stability (slope of the signal over 24 h) were calculated to study both random fluctuations and long-term baseline stability, respectively. All IMC experiments were performed in biological triplicates (with each of these biological replicates consisting of a technical duplicate). A biological replicate was considered positive if one of two technical replicates exceeded the 1 µW detection threshold.

原文截图 · PDF p.3, Methods, baseline and positivity rule

热力学参照

本文使用生理盐水参照容器，辅助区分仪器/环境热背景。参照不等同于装有培养基的阴性过程对照。

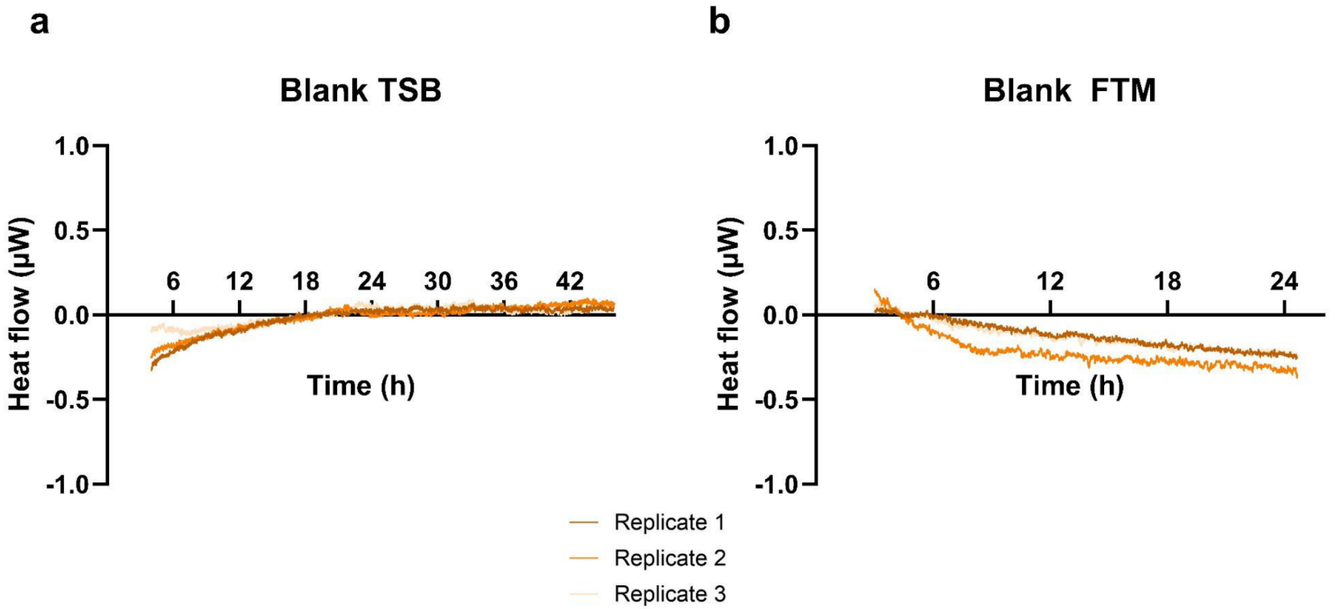
未接种培养基空白

TSB与FTM各3个重复，用来评估基线噪声及漂移：1 h区间SD表示短期随机波动；24 h信号斜率表示长期漂移。

空白基线验证支持阈值选择，但不等同于真实药品基质的特异性验证。

Sioen I, Coenye T. J Pharm Innov 22:107 (2027); online 2026-08-24. DOI: 10.1007/s12247-026-10986-0
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Online Resource 2 / Fig.A1：空白热流全图



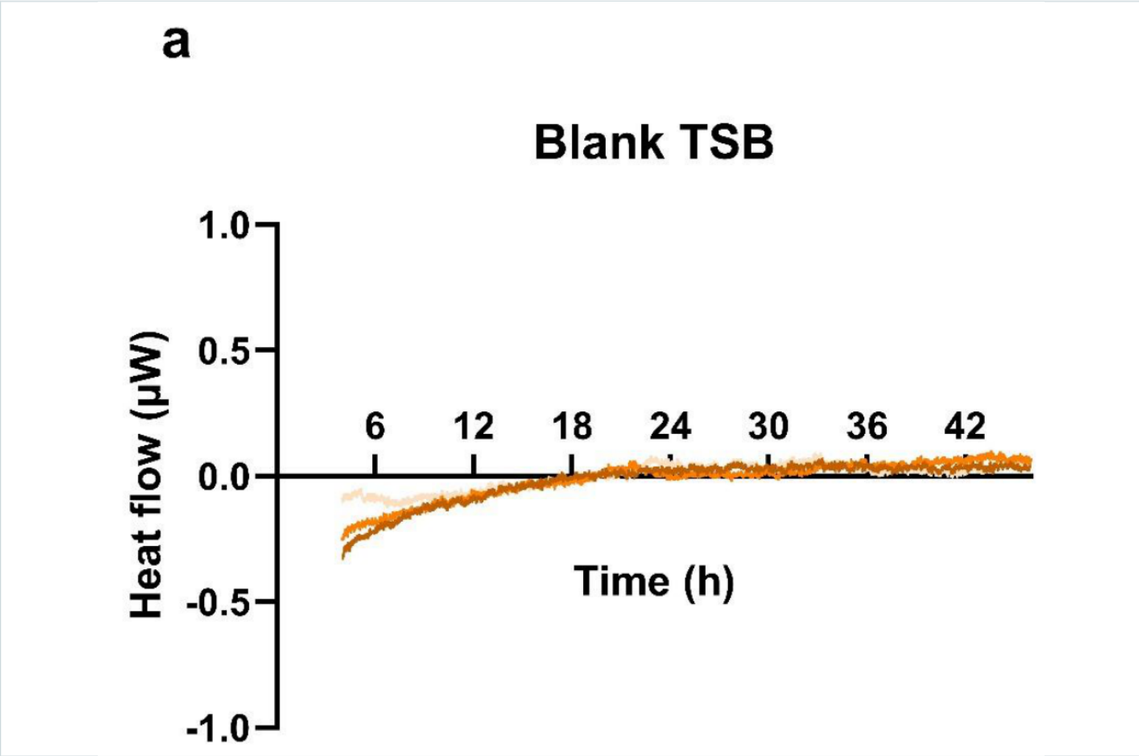
Online Resource 2 – Fig. A1 Thermograms of uninoculated samples of (a) TSB and (b) FTM, both in volumes of 400 µL. Short term noise and baseline drift equal 0.019 µW and 0.00069 µW/h for TSB and 0.041 µW and - 0.014 µW/h for FTM.

原文截图 · Supplement PDF p.2, Online Resource 2 Fig.A1a–b complete

两图均为未接种空白，各3条重复曲线。纵轴热流（µW）；横轴时间（h）。图a和图b的时间范围不同，不能用同样的横向长度比较漂移速度。

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Fig.A1a：TSB空白与稳定区间



原文截图 · Supplement PDF p.2, Online Resource 2 Fig.A1a, TSB blank



原文截图 · Supplement PDF p.2, Fig.A1 图例

图形与数值

3条空白曲线起初为负，随后接近零。作者给出短期噪声SD＝0.019 µW；基线漂移＝0.00069 µW/h。

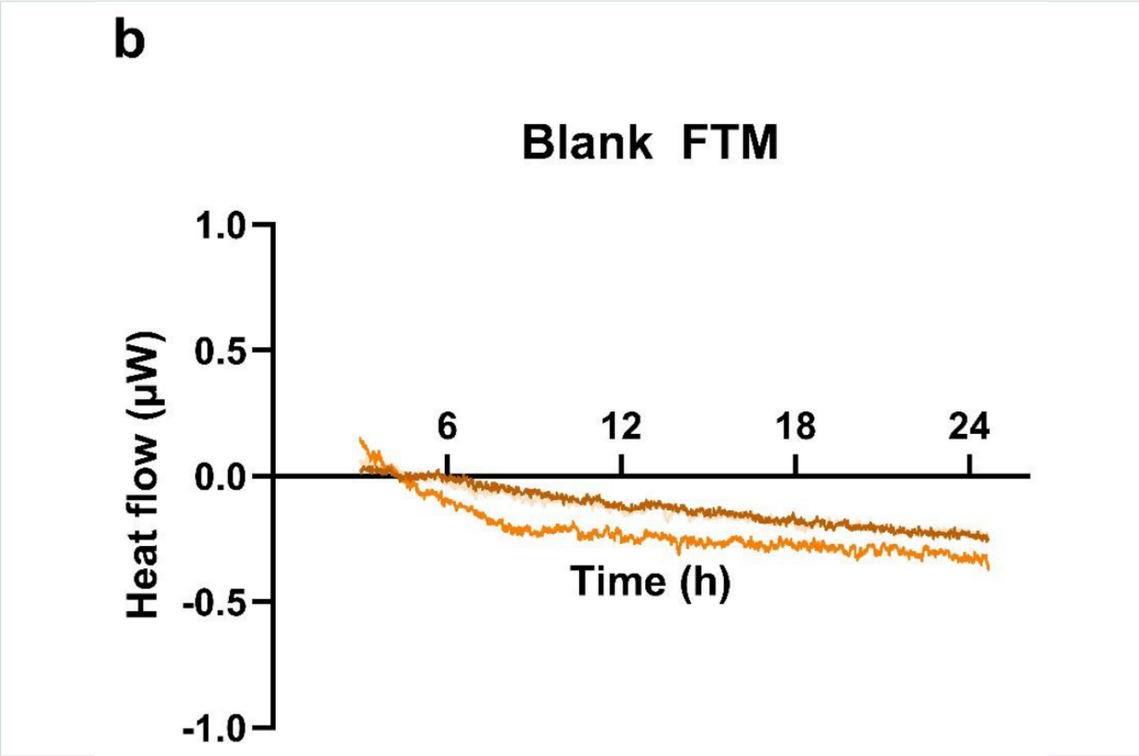
读出含义

负热流可来自相对基线/参照，不等于“负的微生物数量”。起始瞬态与后段近稳态需分开。

噪声SD来自代表性1 h区间，不是整条曲线的SD；图中未标明该区间的位置。

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Fig.A1b：FTM的负向漂移



原文截图 · Supplement PDF p.2, Online Resource 2 Fig.A1b, FTM blank



原文截图 · Supplement PDF p.2, Fig.A1 图例

图形与数值

FTM空白随时间向负值移动。短期噪声SD＝0.041 µW；漂移＝−0.014 µW/h。

据斜率的教学计算

若24 h内保持该线性斜率，变化量约−0.014×24＝−0.336 µW。此为斜率换算，不是新增测量点。

负向漂移不易触发正阈值，却会改变信号距离阈值的余量。跨基质或更长观察期仍需重新评估。

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1 μW阈值的证据与限制

criterion is that the detection threshold should exceed the mean baseline value plus three times the SD. For TSB, mean signal + 3SD equals 0.162 μW; the 1 μW detection threshold approximately equals mean + 14SD. For FTM, mean + 3SD equals 0.164 μW; 1 μW approximately equals mean + 11SD.

原文截图 · PDF p.4, Results, 24 h baseline statistics (text excerpt)

原文的24 h基线比较

TSB：均值+3SD＝0.162 μW；FTM：0.164 μW。1 μW分别约为均值+14SD与均值+11SD。

阈值高于两种已测空白的波动范围，支持降低由这些基线波动造成的误报。

不要混用三种统计量

0.019/0.041 μW是1 h噪声SD；14SD/11SD使用24 h基线统计；漂移单位为μW/h。它们不是同一个SD。

本文没有提供完整基线均值μ、SD σ和逐点数据，不能精确复算所有阈值倍数，也不能把1 μW叫作CFU检出限。

Sioen I, Coenye T. J Pharm Innov 22:107 (2027); online 2026-08-24. DOI: 10.1007/s12247-026-10986-0
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Fig.1：热曲线、阈值与技术重复的全图

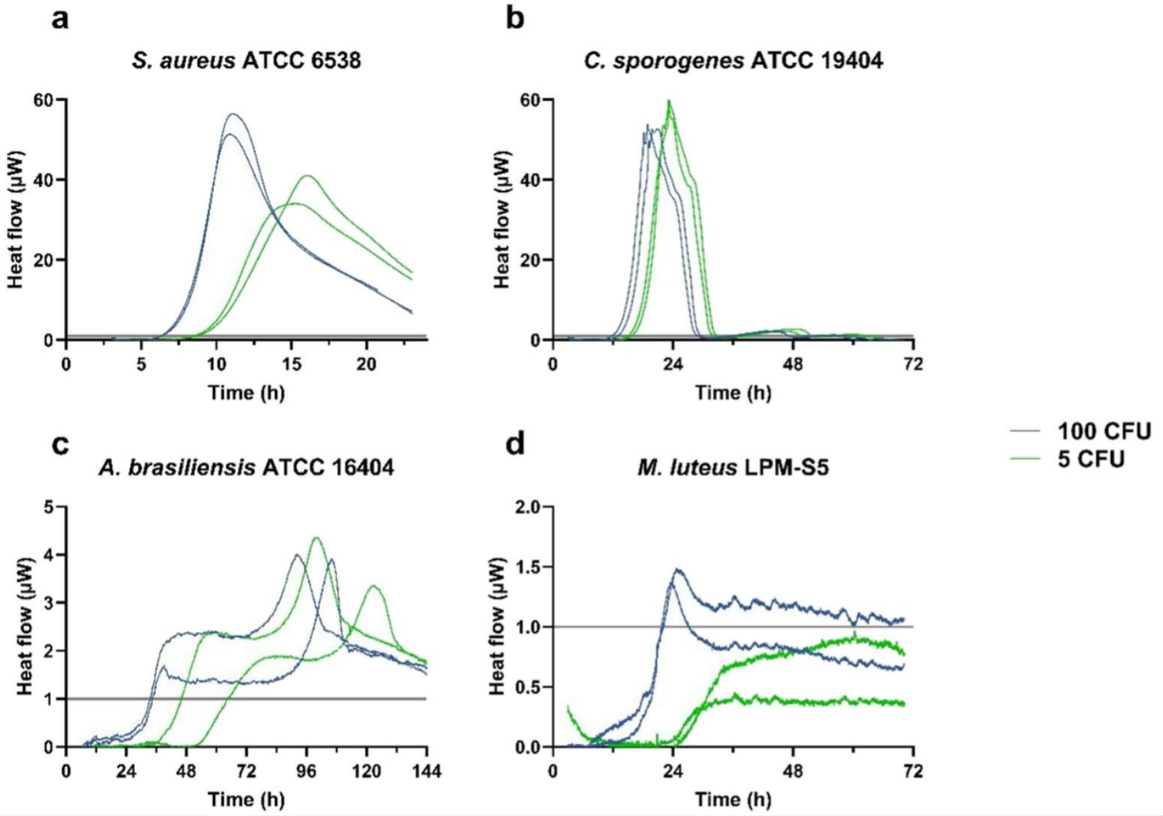


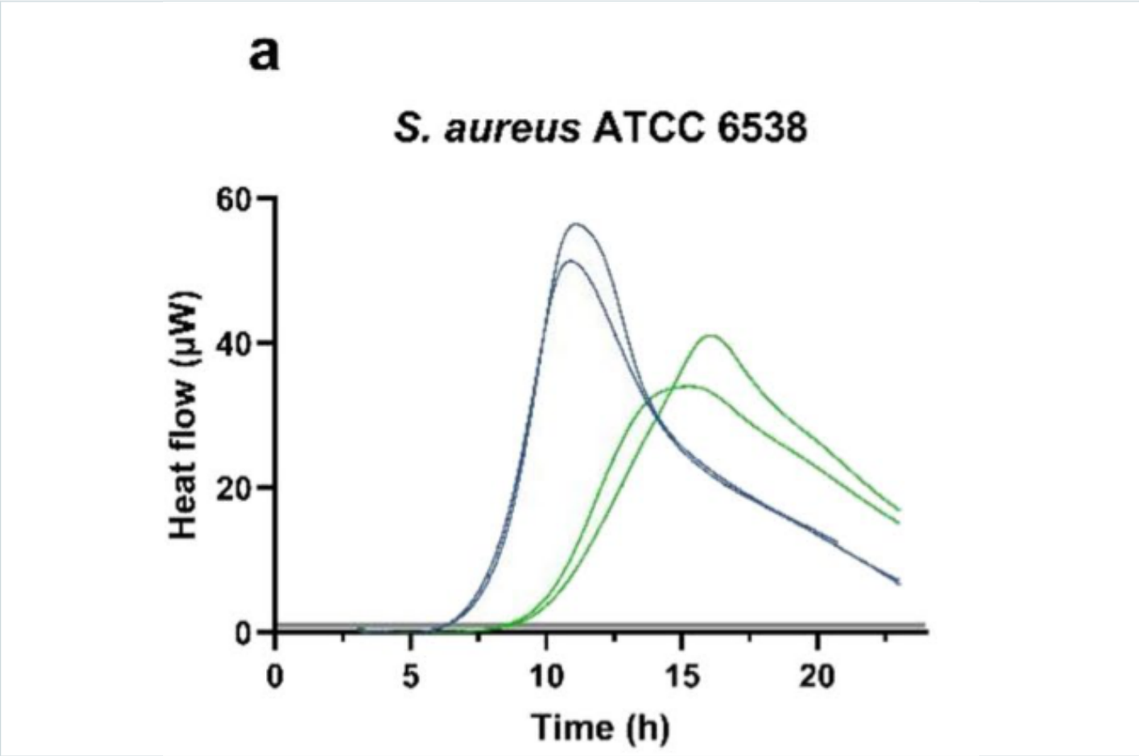
Fig. 1 Thermograms for a biological replicate (consisting of two technical replicates) of (**a**) *S. aureus* (in FTM), (**b**) *C. sporogenes* (in FTM), (**c**) *A. brasiliensis* (in TSB) and (d) *M. luteus* (in TSB, lower inoculum size undetected) for the theoretical inoculum sizes of 100 CFU and 5 CFU. The grey line at 1 µW indicates the detection threshold

原文截图 · PDF p.5, Fig.1a–d complete with caption and legend

a–d各表示一个生物学重复的两支技术重复。蓝线为名义100 CFU，绿线为5 CFU；灰色1 µW线为阈值。各子图纵横轴范围不同，必须逐图读。

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Fig.1a：S. aureus的起升与峰值



原文截图 · PDF p.5, Fig.1a (共享图例见下)



原文截图 · PDF p.5, Fig.1 共享图例

曲线怎样变成判定

蓝绿曲线随时间起升并过阈值。TTD取首次过1 µW的时刻，不取峰值时刻或最大热流。

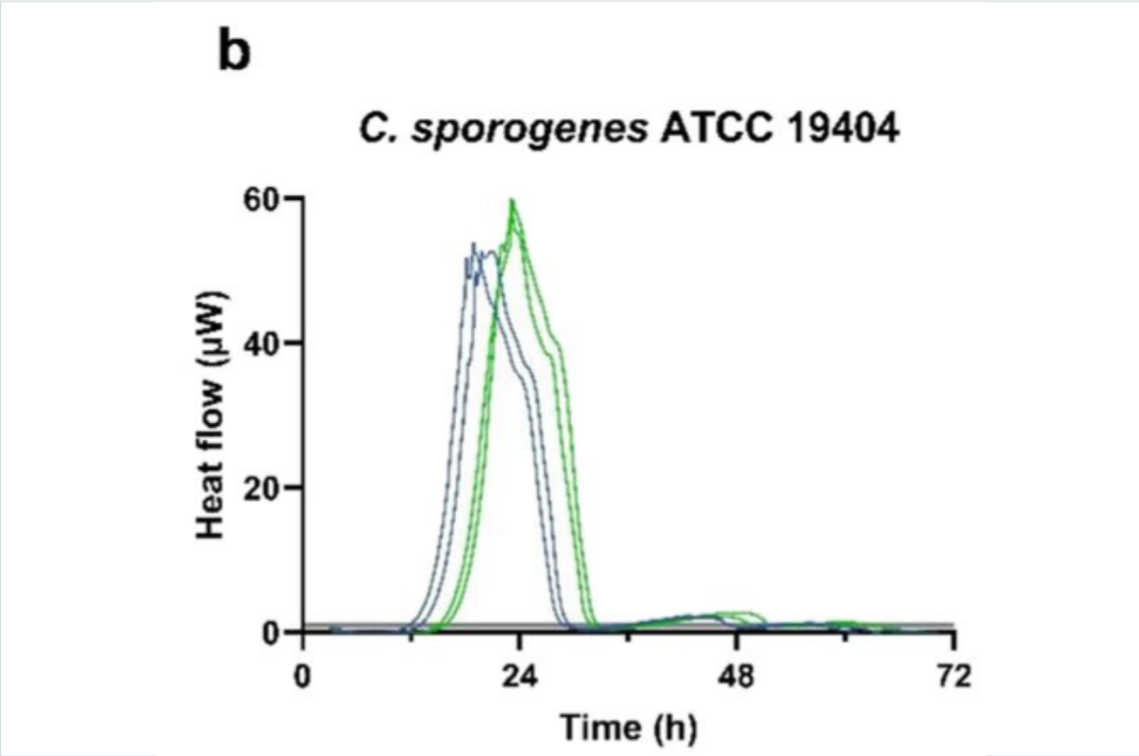
100 CFU曲线较早起升；峰后下降仍然是已经报阳的样品，不能因末端回落而改判阴性。

表示范围

这里是一个生物学重复的代表曲线，不是3个生物学重复的均值。两条同色曲线反映技术重复。

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Fig.1b：C. sporogenes的窄热流峰



原文截图 · PDF p.5, Fig.1b (共享图例见下)



原文截图 · PDF p.5, Fig.1 共享图例

窄峰与连续记录

该子图的热流集中在较窄时间段，随后接近基线。一次末端读数可能遗漏已经出现的热流峰；连续记录保留其出现历史。

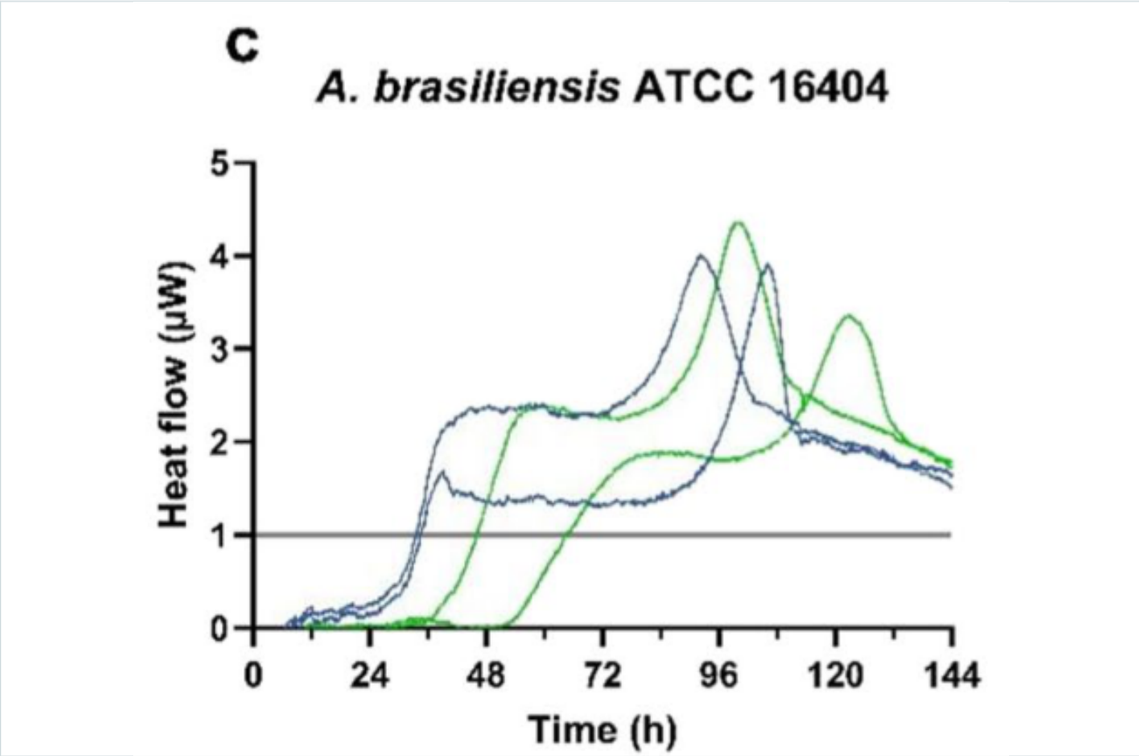
峰高不直接代表初始CFU。本文判阳仍只使用1 µW门槛，而不是按峰形分类。

代表图的边界

该代表重复不能证明所有技术支都检出。Online Resource 3显示高水平此菌的支级检出为5/6。

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Fig.1c：A. brasiliensis的多阶段热流



原文截图 · PDF p.5, Fig.1c (共享图例见下)



原文截图 · PDF p.5, Fig.1 共享图例

多峰不等于多菌种

曲线包含多个上升阶段和局部峰。仅由这张图不能把各峰归因为特定代谢通路或多个物种。

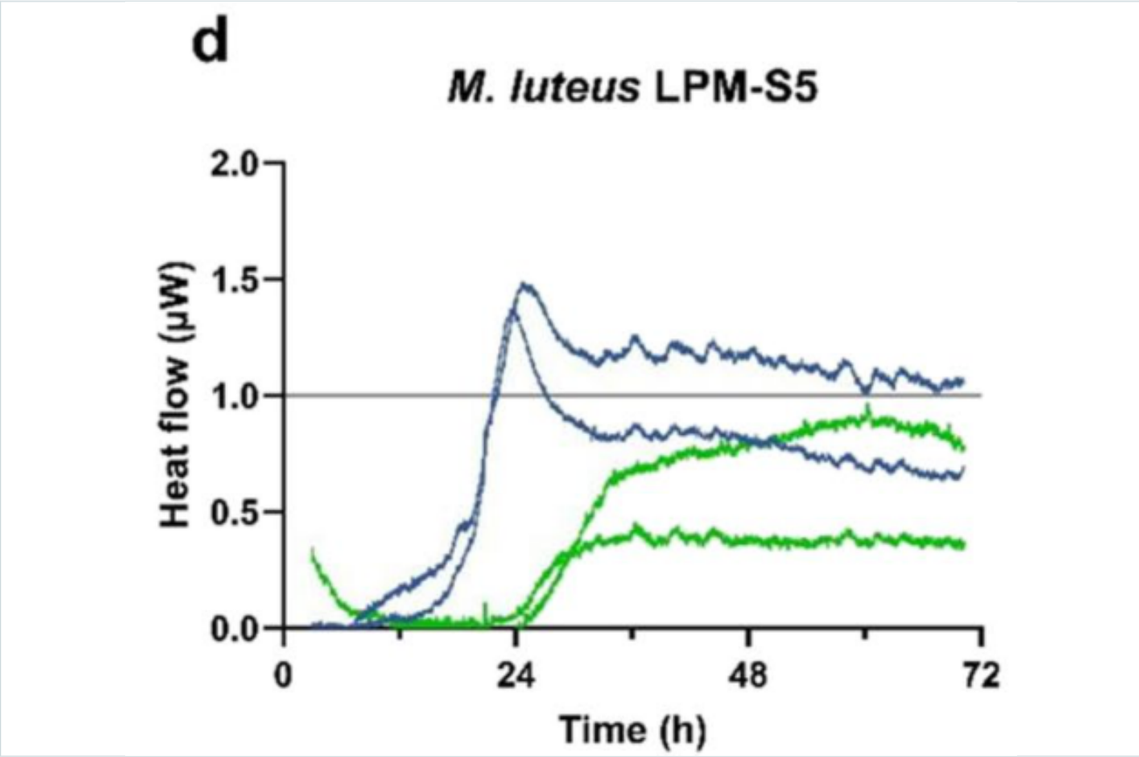
横轴延伸至144 h，纵轴仅约数µW；不能与a、b的峰高直接比较初始微生物数量。

检出与持续表征

阈值过线时即形成TTD；后续热流形状可以保留，但其机制解释不属于本研究的验证结论。

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Fig.1d：有热流上升却仍判为未检出



原文截图 · PDF p.5, Fig.1d (共享图例见下)



原文截图 · PDF p.5, Fig.1 共享图例

必须保留的失败结果

M. luteus低水平绿线有所上升，却未超过灰色1 µW阈值，因此该生物学重复仍为未检出。

“看见变化”和“达到预定阳性判据”不同；不能在看图之后为个别样品临时放宽阈值。

原因尚未验证

作者提出容器内可用氧不足的可能解释，未直接验证。这里不把假设写成确定原因，也不提供培养优化操作。

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Table 2：报阳比例的完整分母

Table 2 Number of positive cultures detected by the compendial sterility test and the microcalorimetry-based method for the different inoculum sizes

Target inoculum size	Number of positive cultures / total number of cultures tested (%)	
	Compendial method	Microcalorimetry-based method
100 CFU	48/48 (100%)	48/48 (100%)
5 CFU	36/48 (75.0%)	44/48 (91.7%)*
Total	84/96 (87.5%)	92/96 (95.8%)*

* Statistically significant difference compared to compendial method; $p < 0.05$ (Exact McNemar test)

原文截图 · PDF p.4, Table 2 complete

原文结果

高水平：两方法均48/48。低水平：参照36/48＝75.0%，IMC 44/48＝91.7%。合并为84/96与92/96。

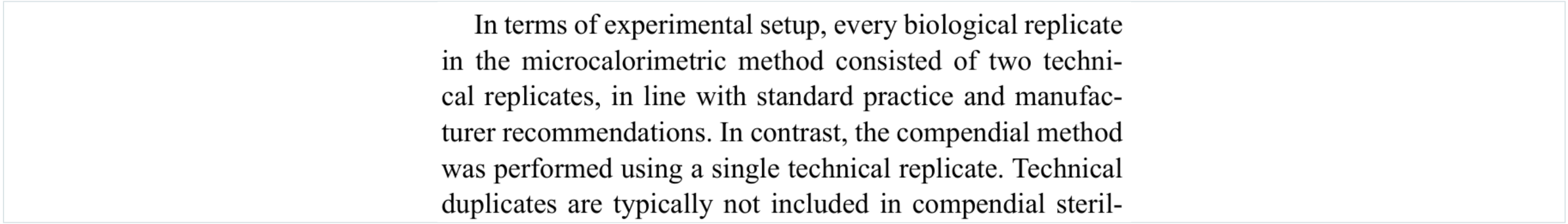
据表计算

低水平增加8/48＝16.7个百分点；合并增加8/96＝8.3个百分点。均不是“提高16.7%灵敏度”。

作者报告Exact McNemar p<0.05；表中未列出完整配对四格表。真实污染状态未独立确认，不能计算可靠灵敏度/特异性。

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技术重复判阳规则会影响检测比例



In terms of experimental setup, every biological replicate in the microcalorimetric method consisted of two technical replicates, in line with standard practice and manufacturer recommendations. In contrast, the compendial method was performed using a single technical replicate. Technical duplicates are typically not included in compendial steril-

原文截图 · PDF p.7, Discussion, technical replicate asymmetry (text excerpt)

教学推断，非论文估计

若单支检出概率为p，两支独立且同分布，则“任一阳性”概率：

$$P(\text{至少一支阳性})=1-(1-p)^2$$

假设p=0.80，则$1-0.20^2=0.96$。该96%只是规则演示，不是本文的性能估计。

成立条件与实际限制

同一生物学重复的两支可能相关，未必独立。若高度相关，重复带来的概率增益会更小。

IMC用两支、参照用一支，可能部分解释报阳数量差异。要评价传感器本身，需把规则效应与热读出效应分开。

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Fig.2：更早报阳的平均时间

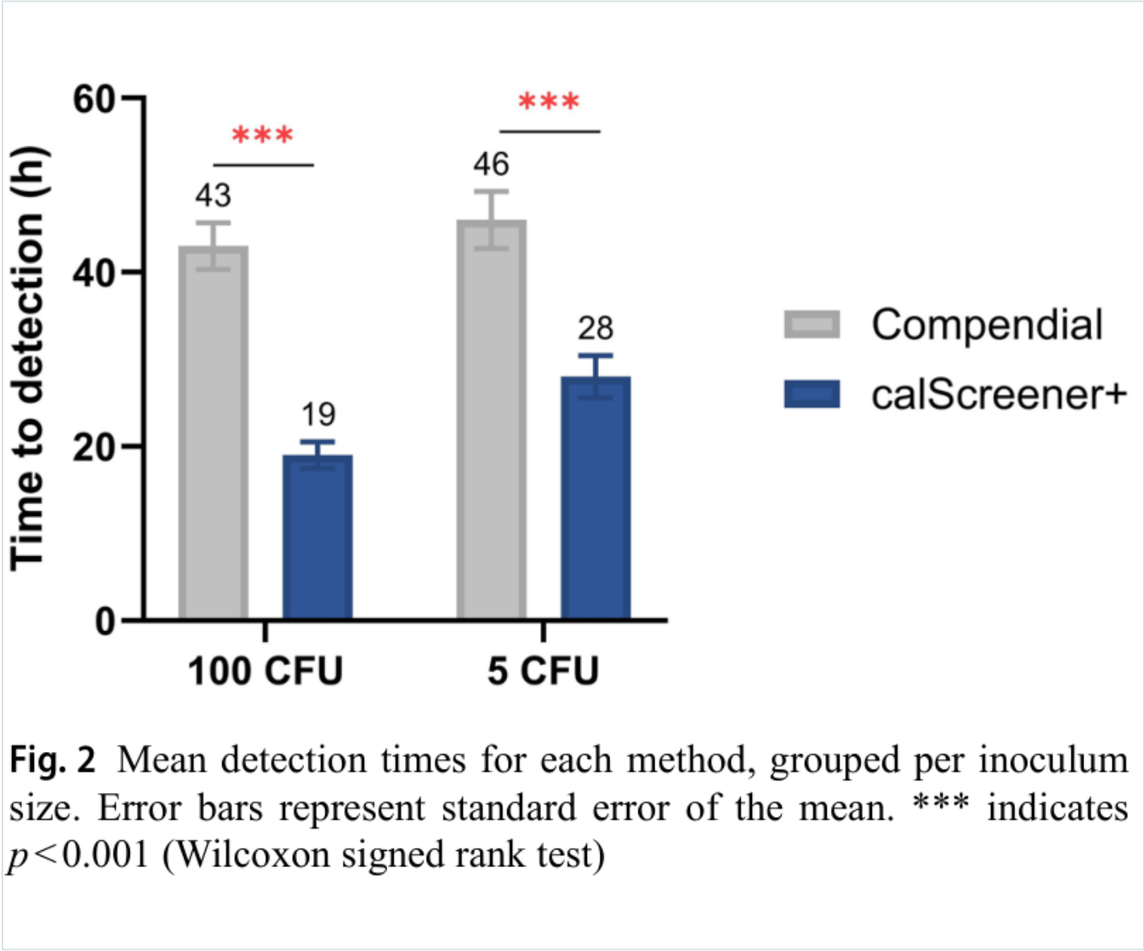


Fig. 2 Mean detection times for each method, grouped per inoculum size. Error bars represent standard error of the mean. *** indicates $p < 0.001$ (Wilcoxon signed rank test)

原文截图 · PDF p.5, Fig.2 complete with caption

原文报告的组均值

名义100 CFU：参照43 h，IMC 19 h，差24 h。名义5 CFU：46 h与28 h，差18 h。

误差线为SEM（均值标准误），不是SD或95%置信区间。作者报告Wilcoxon配对符号秩检验p<0.001。

时间边界

它是检测观察过程中的报阳时间，不是从取样到最终放行的完整周转时间；也不支持所有阴性样品在19 h或28 h放行。

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未检出、右删失与统计比较

Statistical Analysis

The exact McNemar test was used to compare the number of positive cultures detected by the compendial and calScreener+ method for each inoculum concentration. TTD per inoculum size was compared between methods using a Wilcoxon signed rank test. To compare the TTD between methods across organisms, a paired sample t-test was performed after confirming normal distribution of the data (Quantile-Quantile [QQ] plots). For the 100 CFU inoculum size, TTD comparisons per organism were performed with paired sample t-tests. Bonferroni correction was used to adjust for multiple comparisons. All statistical tests were performed using IBM SPSS Statistics (version 29.0.1.0). For all tests, a p-value of <0.05 was considered statistically significant. Replicates not detected within the 336-hour detection window result in right-censored data that cannot be analyzed by a standard parametric test (e.g. t-test). Although survival analysis methods (e.g. Cox regression) can handle censored data, the sample size per condition ($n=3$) in this study does not provide sufficient statistical power. Therefore, statistical analysis was not performed for the 5 CFU inoculum size, as some replicates remained undetected with this inoculum.

原文截图 · PDF p.4, Statistical Analysis

336 h内未过阈值

这类结果为右删失：只知道TTD大于观察窗或未出现可检测信号。不能把它当作TTD=0，也不能当作精确336 h。

原文统计的两个层级

Fig.2报告两档总体Wilcoxon检验；Table 3只对100 CFU逐物种做配对t检验并Bonferroni校正。

方法段“5 CFU未做统计”的宽泛表述与Fig.2低水平p值并列。未说明总体分析如何处理未检出/配对缺失，不能擅自补出算法或复算p值。

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Table 3：逐物种TTD完整表

Table 3 Detection times for each method for each organism. Data are presented as mean±standard deviation (SD, $n=3$). ND, not detected (i.e. all three replicates were negative). Statistics were performed on the 100 CFU inoculum size only (as for these, there are no missing data)

Microorganism	Time to detection (h) (mean±SD)			
	100 CFU		5 CFU	
	Compendial method	Microcalorimetry-based method	Compendial method	Microcalorimetry-based method
A. delafieldii LPM-S1	48±0	26.0±1.9*	48±0[a]	36.8±2.8
A. brasiliensis ATCC 16404	68±0	35.1±1.1*	75.7±13.3	49.3±5.5
A. versicolor LPM-S2	72±0	30.1±1.4*	72±0[a]	52.9±4.8
B. spizizenii ATCC 6633	48±0	16.1±0.3*	ND	18.7±0.3[a]
B. subtilis LPM-S3	48±0	16.6±1.3*	ND	19.9±1.8[a]
B. contaminans LPM-S4	46±0	17.0±0.1*	46±0	21.5±0.3
C. albicans ATCC 10231	56.7±12.2	12.1±1.6	70±0	19.7±5.1
C. sporogenes ATCC 19404	22±0	13.9±1.3	22±0	15.2±0.3
C. acnes LMG 16711	46±0	41.4±1.2	70±0	49.3±4.0
M. luteus LPM-S5	59±0	22.1±0.4*	59±0	57.3±3.4[a]
P. chrysogenum LPM-S6	68±0	33.1±0.8*	68±0[a]	44.6±2.3
P. paraeruginosa ATCC 9027	22±0	8.6±1.0*	22±0	10.4±1.0
S. aureus ATCC 6538	22±0	6.6±0.2*	22±0	9.1±0.6
S. epidermidis LPM-S7	22±0	11.5±0.2*	22±0[a]	13.9±1.0
S. hominis LPM-S8	22±0	9.5±0.4*	22±0[a]	18.0±7.8[a]
S. pyogenes LMG 14237	22±0	9.7±0.2*	22±0[a]	12.8±1.4

* Statistically significant difference compared to compendial method; $p<0.05$ (paired sample t-test with Bonferroni correction)

[a] Not all three replicates were detected; undetected replicates were not taken into account for calculation of the mean. Figure 3 displays how many replicates were undetected for each method

原文截图 · PDF p.6, Table 3 complete

此表误差为SD；Fig.2/3误差为SEM，两者不能混称。a标记的低水平均值排除了未检出重复，实际有效n小于3。完整表后分上下半放大。

Sioen I, Coenye T. J Pharm Innov 22:107 (2027); online 2026-08-24. DOI: 10.1007/s12247-026-10986-0
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Table 3上半：时间差与非显著结果

Microorganism	Time to detection (h) (mean ± SD)			
	100 CFU		5 CFU	
	Compendial method	Microcalorimetry-based method	Compendial method	Microcalorimetry-based method
A. delafieldii LPM-S1	48 ± 0	26.0 ± 1.9*	48 ± 0[a]	36.8 ± 2.8
A. brasiliensis ATCC 16404	68 ± 0	35.1 ± 1.1*	75.7 ± 13.3	49.3 ± 5.5
A. versicolor LPM-S2	72 ± 0	30.1 ± 1.4*	72 ± 0[a]	52.9 ± 4.8
B. spizizenii ATCC 6633	48 ± 0	16.1 ± 0.3*	ND	18.7 ± 0.3[a]
B. subtilis LPM-S3	48 ± 0	16.6 ± 1.3*	ND	19.9 ± 1.8[a]
B. contaminans LPM-S4	46 ± 0	17.0 ± 0.1*	46 ± 0	21.5 ± 0.3
C. albicans ATCC 10231	56.7 ± 12.2	12.1 ± 1.6	70 ± 0	19.7 ± 5.1
C. sporogenes ATCC 19404	22 ± 0	13.9 ± 1.3	22 ± 0	15.2 ± 0.3

原文截图 • PDF p.6, Table 3 column headings and rows 1–8

前6项高水平IMC均值都较低且带*；C. albicans为56.7→12.1 h，C. sporogenes为22→13.9 h，但无*。未经校正的“大差值”不等于校正后显著。低水平两种Bacillus参照均ND。

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Table 3下半：最快与慢检出的物种

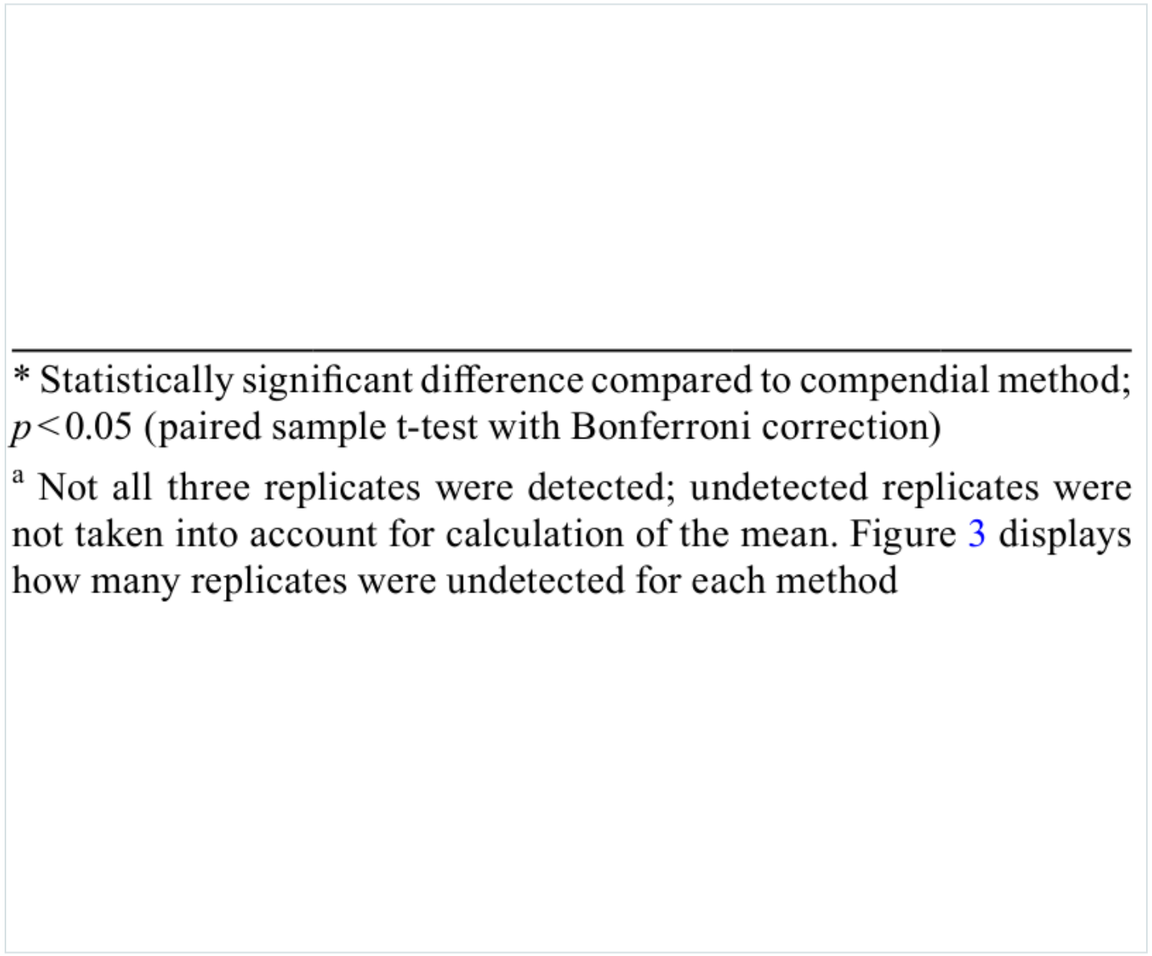
C. acnes LMG 16711	46±0	41.4±1.2	70±0	49.3±4.0
M. luteus LPM-S5	59±0	22.1±0.4*	59±0	57.3±3.4[a]
P. chrysogenum LPM-S6	68±0	33.1±0.8*	68±0[a]	44.6±2.3
P. paraeruginosa ATCC 9027	22±0	8.6±1.0*	22±0	10.4±1.0
S. aureus ATCC 6538	22±0	6.6±0.2*	22±0	9.1±0.6
S. epidermidis LPM-S7	22±0	11.5±0.2*	22±0[a]	13.9±1.0
S. hominis LPM-S8	22±0	9.5±0.4*	22±0[a]	18.0±7.8[a]
S. pyogenes LMG 14237	22±0	9.7±0.2*	22±0[a]	12.8±1.4

原文截图 · PDF p.6, Table 3 rows 9–16

四列顺序为高水平参照/IMC、低水平参照/IMC，单位h。C. acnes高水平46→41.4 h，无*；其余下半表条目高水平均带*。S. aureus均值6.6 h较早，M. luteus低水平57.3 h仅来自已检出重复。

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Table 3脚注：条件均值不代表全部样品



* Statistically significant difference compared to compendial method; $p < 0.05$ (paired sample t-test with Bonferroni correction)

[a] Not all three replicates were detected; undetected replicates were not taken into account for calculation of the mean. Figure 3 displays how many replicates were undetected for each method

原文截图 · PDF p.6, Table 3 statistical and missing-data footnotes

a脚注改变了分母

作者排除未检出重复后计算均值。低水平Bacillus的18.7 h/19.9 h，只描述两次检出；第三次未检出仍需单独保留。

显著性与等效性

星号仅标识高水平经Bonferroni校正后p<0.05。C. albicans、C. sporogenes和C. acnes无星号，不表示与参照等效。

未报告效应量置信区间或预设等效界值；不能用“无显著差异”宣称等效验证通过。

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Fig.3：全部生物学重复与ND位置

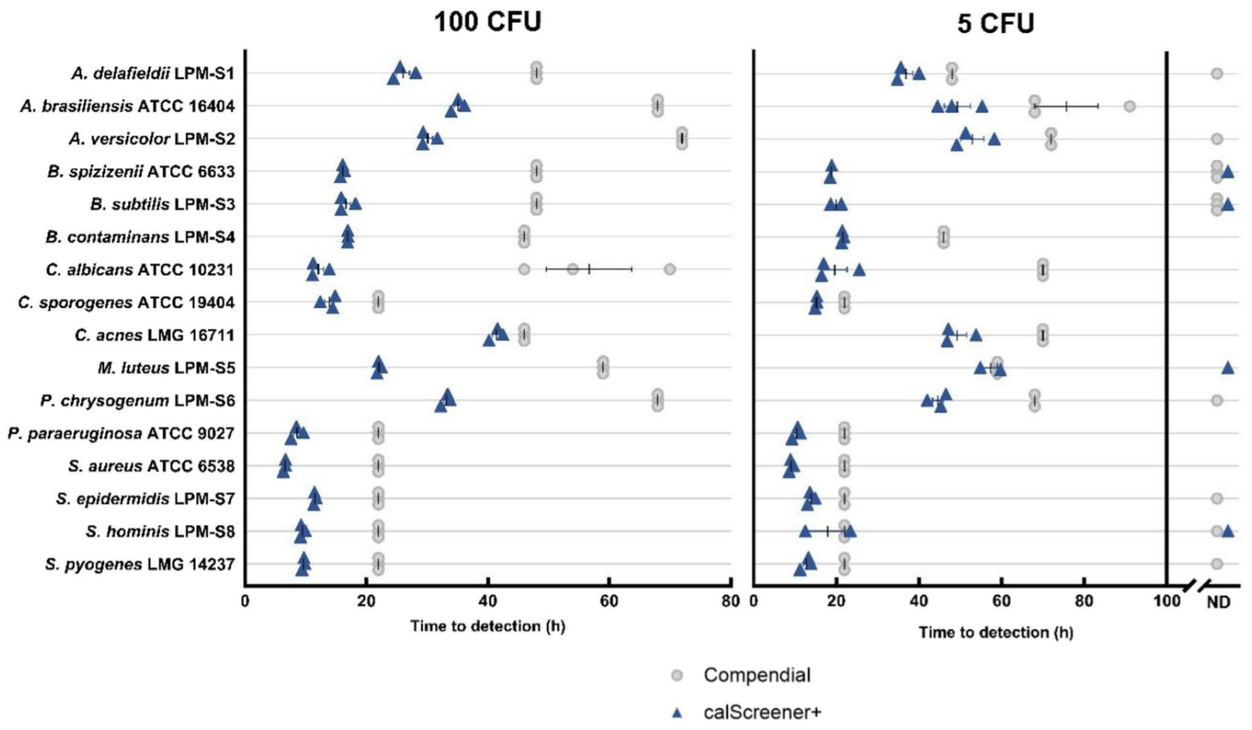


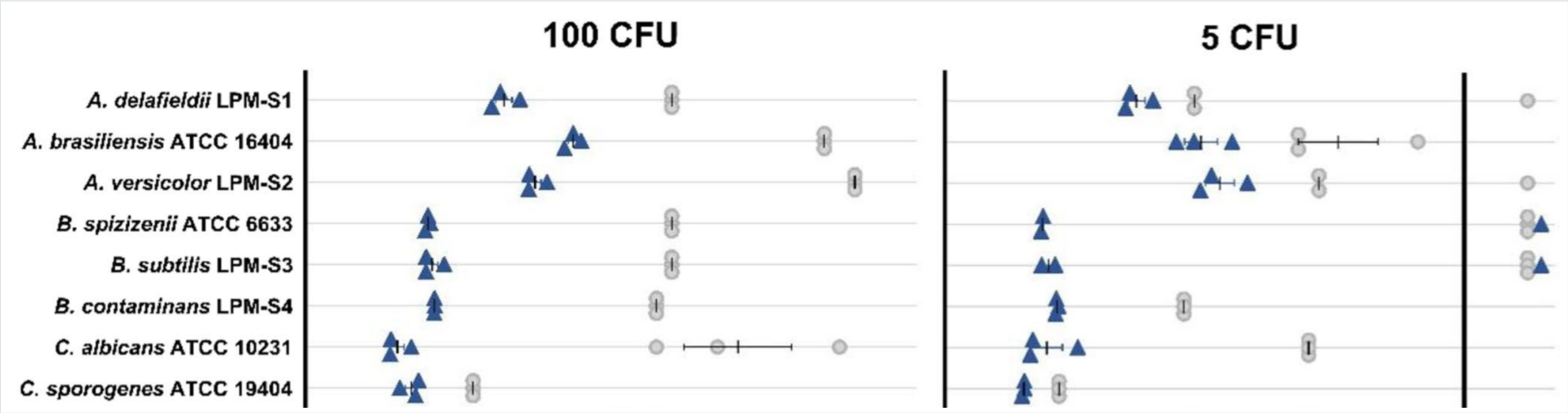
Fig. 3 Time to detection for the individual replicates of all tested microorganisms for the 100 CFU inoculum (left) and the 5 CFU inoculum (right). Replicates for which no growth was observed within the 336 h detection window are shown as not detected (ND). Error bars represent mean ± standard error of the mean

原文截图 · PDF p.7, Fig.3 complete with caption and legend

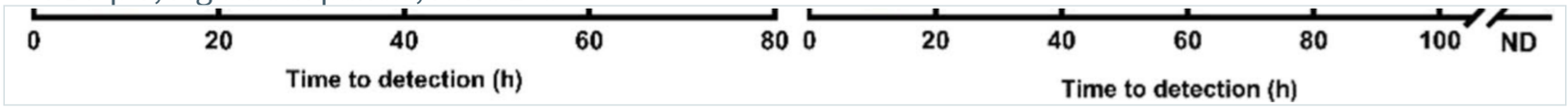
灰圆为参照，蓝三角为IMC。左为100 CFU，右为5 CFU；点表示单个生物学重复，误差线为均值±SEM。ND区域表示336 h内未检出，不是横轴100 h附近的真实TTD。

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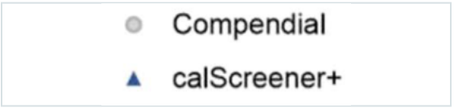
Fig.3前8项：低水平的Bacillus差异



原文截图 · PDF p.7, Fig.3 both panels, rows 1–8



原文截图 · PDF p.7, Fig.3 x axes

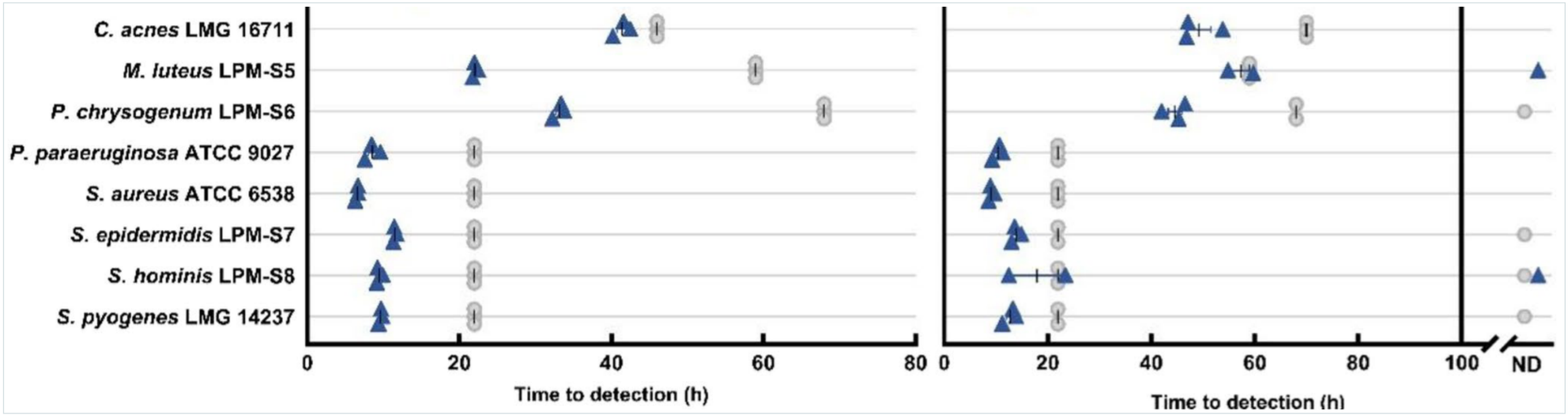


原文截图 · PDF p.7, Fig.3 shared legend

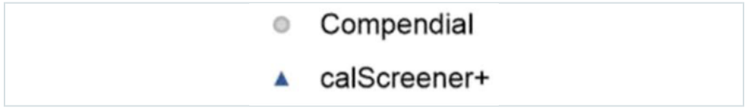
A. delafieldii与A. versicolor各有参照未检出；两种Bacillus参照各0/3、IMC各2/3。A. brasiliensis、B. contaminans、C. albicans、C. sporogenes在低水平两方法均检出3/3。该结果仍受双技术重复规则影响。

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Fig.3后8项：未检出不能被均值隐藏



原文截图 · PDF p.7, Fig.3 both panels, rows 9–16 and x axes

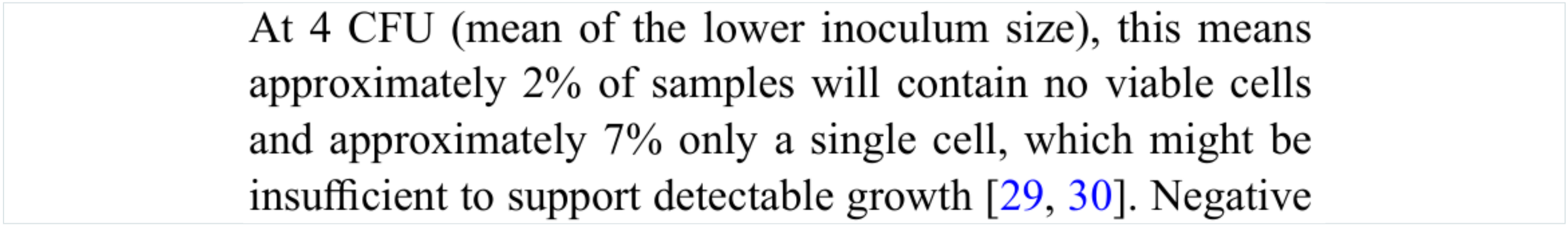


原文截图 · PDF p.7, Fig.3 shared legend

低水平IMC中M. luteus和S. hominis各2/3，其余后半表物种3/3。参照的P. chrysogenum、S. epidermidis、S. hominis和S. pyogenes各有未检出。C. acnes、M. luteus、P. paraeruginosa及S. aureus参照均3/3。

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低挑战量的随机性：Poisson教学演算



At 4 CFU (mean of the lower inoculum size), this means approximately 2% of samples will contain no viable cells and approximately 7% only a single cell, which might be insufficient to support detectable growth [29, 30]. Negative

原文截图 · PDF p.7, Discussion, Poisson example (text excerpt)

从论文讨论展开的计算

$P(N=k)=e^{(-\lambda)}\lambda^k/k!$

λ为平均单位数，N为一次样品中的实际单位数，k为0、1、2…；均无量纲。e为自然常数，k!为阶乘，0!=1。若λ=4：

$P(N=0)=e^{-4}\approx 1.83\%$

$P(N=1)=4e^{-4}\approx 7.33\%$

模型可以解释多少

原文近似表述为2%空样品、7%仅一个单位。成立条件是随机、独立、均匀分配；聚集等情况会偏离Poisson模型。

低水平两方法合并未检出16/96＝16.7%。这不是1.83%的直接验证，也不能全部归因为抽样随机性；热阈值、输入与检测条件仍可能影响。

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精密度计算：同次重复与跨次重复

Repeatability was quantified as the intra-assay percent relative SD (%RSD), determined by expressing the SD calculated between the TTD of technical duplicates relative to their mean. The %RSD was calculated for each biological replicate and averaged to determine the mean intra-assay %RSD per organism. For the ruggedness, the SD between

原文截图 · PDF p.3, Methods, RSD definition (text excerpt)

原文定义

$$\%RSD = 100 \times s / \bar{t}$$

t̄为TTD均值（h），s为TTD样本SD（h）。技术双重复内先计算RSD，再在生物学重复间平均，得到repeatability。

假设算例与边界

假设两技术TTD为18、20 h：t̄=19 h；样本s=√2≈1.414 h；RSD≈7.44%。

跨生物学重复的TTD计算RSD得到本文ruggedness。它不代表跨实验室、操作者或设备的完整耐变性评价。未检出会减少可计算的数据对。

Sioen I, Coenye T. J Pharm Innov 22:107 (2027); online 2026-08-24. DOI: 10.1007/s12247-026-10986-0
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Online Resource 3：重复性完整表

Online Resource 3 – Table Repeatability assessment of the calScreener+ sterility testing method. Results are based on three biological replicates, each consisting of two technical replicates (N = 6). Detection rate indicates the number of positive replicates (n) out of all tested vials. Mean relative standard deviation (%RSD) values, calculated based on the time to detection between technical replicates, are reported per organism, together with the range across biological replicates.

	Inoculum size			
	100 CFU		5 CFU	
Organism	Mean %RSD (range)	Detection rate (n/6)	Mean %RSD (range)	Detection rate (n/6)
***Acidovorax delafieldii* LPM-S1**	2.71% (1.03 – 4.58%)	6/6	12.04% (2.55 – 21.54%)[a]	5/6
***Aspergillus brasiliensis* ATCC 16404**	3.35% (2.03 – 5.89%)	6/6	23.75% (N/A)[b]	4/6
***Aspergillus versicolor* LPM-S2**	1.61% (0.24 – 2.32%)	6/6	8.34% (6.29 – 10.39%)[a]	5/6
***Bacillus spizizenii* ATCC 6633**	0.63% (0.39 – 0.97%)	6/6	N/A[c]	2/6
***Bacillus subtilis* LPM-S3**	1.35% (0.38 – 2.56%)	6/6	N/A[c]	2/6
***Burkholderia contaminans* LPM-S4**	0.60% (0.30 – 1.97%)	6/6	1.43% (1.08 – 1.77%)[a]	5/6
***Candida albicans* ATCC 10231**	0.55% (0.21 – 1.14%)	6/6	0.21% (0.18 – 0.23%)[a]	5/6
***Clostridium sporogenes* ATCC 19404**	19.54% (6.21 – 32.86%)	5/6	3.26% (2.02 – 4.55%)	6/6
***Cutibacterium acnes* LMG 16711**	2.40% (0.23 – 5.34%)	6/6	4.55% (0.69 – 9.43%)	6/6
***Micrococcus luteus* LPM-S5**	1.32% (0.17 – 2.47%)[a]	5/6	24.23% (N/A)[b]	3/6
***Penicillium chrysogenum* LPM-S6**	1.65% (0.42 – 2.76%)	6/6	3.50% (0.27 – 9.69%)	6/6
***Pseudomonas paraeruginosa* ATCC 9027**	9.95% (2.98 – 21.74%)	6/6	3.03% (2.46 – 3.99%)	6/6
***Staphylococcus aureus* ATCC 6538**	3.79% (0.43 – 6.64%)	6/6	4.26% (1.12 – 9.78%)	6/6
***Staphylococcus epidermidis* LPM-S7**	4.31% (1.20 – 7.24%)	6/6	7.26% (4.29 – 11.61%)	6/6
***Staphylococcus hominis* LPM-S8**	4.61% (3.17 – 6.48%)	6/6	42.32% (15.96 – 68.67%)[d]	4/6
***Streptococcus pyogenes* LMG 14237**	1.04% (0.12 – 2.85%)	6/6	5.10% (0.59 – 5.64%)	6/6
Overall	3.71%		10.24%	

Abbreviations: %RSD, relative standard deviation; CFU, colony-forming unit; N/A, not applicable

[a] Mean %RSD calculation based on two biological replicates, as for the remaining biological replicate, only one of the two technical replicates was detected

[b] Value equals the %RSD of a single biological replicate. Averaging was not possible because the remaining replicates were either single-positive or not detected

[c] Mean %RSD could not be calculated due to insufficient data: one biological replicate remained undetected; for the other two biological replicates, only one technical replicate was detected

[d] Mean %RSD calculated based on two biological replicates, as one biological replicate remained completely undetected

原文截图 · Supplement PDF p.3, Online Resource 3 complete table and footnotes

高水平总体平均3.71%，低水平10.24%。表中Detection rate是技术支级n/6，而Table 2按生物学结果计数：每水平n/48、合并n/96；逐物种才是n/3。a–d解释不同缺失情况，后页逐项读。

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Online Resource 3前8项：缺失导致N/A

	Inoculum size			
	100 CFU		5 CFU	
Organism	Mean %RSD (range)	Detection rate (n/6)	Mean %RSD (range)	Detection rate (n/6)
***Acidovorax delafieldii* LPM-S1**	2.71% (1.03 – 4.58%)	6/6	12.04% (2.55 – 21.54%)[a]	5/6
***Aspergillus brasiliensis* ATCC 16404**	3.35% (2.03 – 5.89%)	6/6	23.75% (N/A)[b]	4/6
***Aspergillus versicolor* LPM-S2**	1.61% (0.24 – 2.32%)	6/6	8.34% (6.29 – 10.39%)[a]	5/6
***Bacillus spizizenii* ATCC 6633**	0.63% (0.39 – 0.97%)	6/6	N/A[c]	2/6
***Bacillus subtilis* LPM-S3**	1.35% (0.38 – 2.56%)	6/6	N/A[c]	2/6
***Burkholderia contaminans* LPM-S4**	0.60% (0.30 – 1.97%)	6/6	1.43% (1.08 – 1.77%)[a]	5/6
***Candida albicans* ATCC 10231**	0.55% (0.21 – 1.14%)	6/6	0.21% (0.18 – 0.23%)[a]	5/6
***Clostridium sporogenes* ATCC 19404**	19.54% (6.21 – 32.86%)	5/6	3.26% (2.02 – 4.55%)	6/6

原文截图 · Supplement PDF p.3, Online Resource 3 headings and rows 1–8

两种Bacillus低水平各2/6，但分属两个生物学重复，每次仅一支阳性，所以无法计算成对RSD。A. brasiliensis的23.75%仅来自一个可配对的生物学重复。C. sporogenes高水平5/6，低水平6/6。

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Online Resource 3后8项：平均值掩盖离散

***Cutibacterium acnes* LMG 16711**	2.40% (0.23 – 5.34%)	6/6	4.55% (0.69 – 9.43%)	6/6
***Micrococcus luteus* LPM-S5**	1.32% (0.17 – 2.47%)[a]	5/6	24.23% (N/A)[b]	3/6
***Penicillium chrysogenum* LPM-S6**	1.65% (0.42 – 2.76%)	6/6	3.50% (0.27 – 9.69%)	6/6
***Pseudomonas paraeruginosa* ATCC 9027**	9.95% (2.98 – 21.74%)	6/6	3.03% (2.46 – 3.99%)	6/6
***Staphylococcus aureus* ATCC 6538**	3.79% (0.43 – 6.64%)	6/6	4.26% (1.12 – 9.78%)	6/6
***Staphylococcus epidermidis* LPM-S7**	4.31% (1.20 – 7.24%)	6/6	7.26% (4.29 – 11.61%)	6/6
***Staphylococcus hominis* LPM-S8**	4.61% (3.17 – 6.48%)	6/6	42.32% (15.96 – 68.67%)[d]	4/6
***Streptococcus pyogenes* LMG 14237**	1.04% (0.12 – 2.85%)	6/6	5.10% (0.59 – 5.64%)	6/6
Overall	3.71%		10.24%	

原文截图 · Supplement PDF p.3, Online Resource 3后8项及Overall；列序：100 CFU的RSD、n/6；5 CFU的RSD、n/6

C. acnes、P. chrysogenum、P. paraeruginosa、S. aureus、S. epidermidis和S. pyogenes低水平为6/6。M. luteus为3/6，仅一个成对RSD；S. hominis为4/6，平均42.32%、范围15.96–68.67%，不可用总体10.24%替代。

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Online Resource 3脚注与原表不一致

Abbreviations: %RSD, relative standard deviation; CFU, colony-forming unit; N/A, not applicable

[a] Mean %RSD calculation based on two biological replicates, as for the remaining biological replicate, only one of the two technical replicates was detected
[b] Value equals the %RSD of a single biological replicate. Averaging was not possible because the remaining replicates were either single-positive or not detected
[c] Mean %RSD could not be calculated due to insufficient data: one biological replicate remained undetected; for the other two biological replicates, only one technical replicate was detected
[d] Mean %RSD calculated based on two biological replicates, as one biological replicate remained completely undetected

原文截图 · Supplement PDF p.3, Online Resource 3 footnotes a–d

缺失处理规则

a：只用2个生物学重复；b：仅1个可配对重复；c：没有可配对重复，不能计算；d：1个生物学重复全阴，只用另2个。N/A不是0%。

原表内部核查，非擅自更正

B. contaminans高水平均值写0.60%，范围0.30–1.97%。若按3次RSD的普通均值，最小可能均值为(0.30+0.30+1.97)/3≈0.86%，不能为0.60%。

保留原数值并标注待作者澄清；C. sporogenes高水平5/6却没有相应缺失脚注。无原始TTD，不能自行修正。

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补表算术：另一处待作者澄清的数值

***Streptococcus pyogenes* LMG 14237**	1.04% (0.12 – 2.85%)	6/6	5.10% (0.59 – 5.64%)	6/6
Overall	3.71%		10.24%	

原文截图 · Supplement PDF p.3, Online Resource 3 S. pyogenes and Overall; columns: 100 CFU RSD,n/6; 5 CFU RSD,n/6

S. pyogenes低水平

补表给出的平均RSD为5.10%，范围0.59–5.64%，该组6/6支检出，按方法应有3个成对RSD。

三个值中一个为0.59%，其余不大于5.64%，普通均值至多(0.59+5.64+5.64)/3≈3.96%，不能达到5.10%。

保留原始报告，停止过度复算

不以猜测替换原数值。总体10.24%是作者报告；本课件不以有矛盾的汇总表重建精密度原始数据。仍需原始TTD或作者说明才能解决。

Sioen I, Coenye T. J Pharm Innov 22:107 (2027); online 2026-08-24. DOI: 10.1007/s12247-026-10986-0
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Online Resource 4：跨生物学重复的RSD

Online Resource 4 – Table Ruggedness assessment of the calScreener+ sterility testing method. Relative standard deviation (%RSD) values are calculated based on the time to detection between biological replicates (n = 3).

	Inoculum size	
	100 CFU	5 CFU
Organism	%RSD	%RSD
***Acidovorax delafieldii* LPM-S1**	7.25%	7.68%
***Aspergillus brasiliensis* ATCC 16404**	3.16%	11.17%
***Aspergillus versicolor* LPM-S2**	4.54%	8.99%
***Bacillus spizizenii* ATCC 6633**	1.96%	1.67%[a]
***Bacillus subtilis* LPM-S3**	8.07%	9.02%[a]
***Burkholderia contaminans* LPM-S4**	0.15%	1.33%
***Candida albicans* ATCC 10231**	12.98%	26.09%
***Clostridium sporogenes* ATCC 19404**	9.38%	1.73%
***Cutibacterium acnes* LMG 16711**	2.80%	8.03%
***Micrococcus luteus* LPM-S5**	1.69%	5.99%[a]
***Penicillium chrysogenum* LPM-S6**	2.34%	5.21%
***Pseudomonas paraeruginosa* ATCC 9027**	11.79%	10.01%
***Staphylococcus aureus* ATCC 6538**	3.27%	6.23%
***Staphylococcus epidermidis* LPM-S7**	1.89%	7.03%
***Staphylococcus hominis* LPM-S8**	4.66%	43.15%[a]
***Streptococcus pyogenes* LMG 14237**	2.41%	10.64%
Overall mean	4.89%	10.95%

[a] %RSD calculation based on two biological replicates, as one biological replicate remained undetected

原文截图 · Supplement PDF p.4, Online Resource 4 complete table and footnote

完整表：高水平总体4.89%，低水平10.95%。高水平C. albicans为12.98%；低水平S. hominis为43.15%。a标记只用2个生物学重复；较低RSD仍可能与漏检并存。

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精密度的范围与未检出选择

	Inoculum size	
	100 CFU	5 CFU
Organism	%RSD	%RSD
Acidovorax delafieldii LPM-S1	7.25%	7.68%
Aspergillus brasiliensis ATCC 16404	3.16%	11.17%
Aspergillus versicolor LPM-S2	4.54%	8.99%
Bacillus spizizenii ATCC 6633	1.96%	1.67%[a]
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Streptococcus pyogenes LMG 14237	2.41%	10.64%
Overall mean	4.89%	10.95%

原文截图 · Supplement PDF p.4, Online Resource 4 complete table body

不能仅以总体平均评价设备

高水平重复性范围0.55–19.54%，跨次0.15–12.98%；低水平重复性0.21–42.32%，跨次1.33–43.15%。

两种Bacillus低水平跨次RSD仍可计算，但只基于2次阳性；它不补回第3次未检出的证据。

作者评价与证据边界

作者称总体精密度可接受，同时要求更多重复与后续验证。本文未给出一套预注册、逐物种统一的接受界值或跨实验室验证。

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比较优势中包含读出频率和体系差异

A limitation of the present study is that, while IMC data were collected continuously in real-time, visual inspection was not continuous, leading to a potential overestimation of the TTD for the compendial test. However, the large and consistent TTD differences observed in our study do suggest actual analytical advantages. Moreover, periodic inspection reflects industrial practice. Continuous monitoring offers a clear advantage in Good Manufacturing Practice (GMP) environments as it enables objective, electronically recorded readouts. These can be automated to trigger real-time alerts, allowing faster corrective and preventive actions in case of contamination.

原文截图 · PDF p.6, Discussion, continuous versus daily inspection

时间比较的混杂

IMC连续读数，参照每日查看。参照实际开始可见浑浊的时刻位于两次观察之间，记录TTD可能偏晚。原文承认这一点。

传感器之外的差异

两方法采用相同名义CFU，但论文讨论中体积为5 mL与400 µL，浓度与容器条件不同。双支判阳规则也不同。

参照温度有0.1°C超出推荐范围的记录；生长支持性未另行评价。现有结果比较的是完整研究体系，不能只归因于热传感器。

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用于introduction的证据定位

Finally, while the present study highlights the potential of IMC, formal regulatory validation according to USP < 1223 > or Ph. Eur. 5.1.6 remains necessary before implementing this method in GMP environments. For a qualitative test such as the sterility test, complete validation requires further assessment of remaining critical analytical parameters such as accuracy, specificity and LOD (alongside the precision parameters discussed above) [32]. Moreover, statistical equivalence against the compendial method must be demonstrated by testing a larger sample size (at least 75 to 100 samples) across multiple inoculum levels. Industry guidance documents such as PDA TR no. 33 can be consulted to structure these evaluations [31].

原文截图 · PDF p.8, Discussion, remaining validation

可由本研究支持

等温微量热法提供非光学、连续的代谢热读出。在本文有限菌种与条件下，它获得较多阳性结果和较短平均TTD。

不能由本研究补出

没有正式LoD、可靠灵敏度/特异性、全流程sample-to-answer时间或早期阴性放行时限；未验证复杂制剂或细胞制品的性能。

摘要的“敏感替代方法”应结合正文自限性解释。该研究是概念验证，未完成正式替代方法或GMP实施验证。

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设备研究支持：热学背景与判定策略

Conclusion

We assessed the performance of the calScreener+IMC device using a diverse panel of microorganisms, consisting of reference strains and field isolates. Mean detection times were consistently lower for IMC than for the compendial USP<71>sterility test and IMC also detected a higher number of positive cultures. Our findings highlight IMC as a promising alternative to the compendial test, with the potential of speeding up batch releases while still detecting low-level contamination. Its continuous real-time readout and non-destructive nature makes it well-suited for implementation in GMP environments. Future research will be needed to validate the method, as well as assess calScreener+ performance in more complex matrices, such as cell-based products and formulations containing antimicrobial excipients.

原文截图 · PDF p.8, Conclusion

作者讨论

作者提出继续评估复杂基质、抗微生物辅料以及完整分析性能。连续电子读出有潜在的自动警报用途，但本文没有验证部署效果。

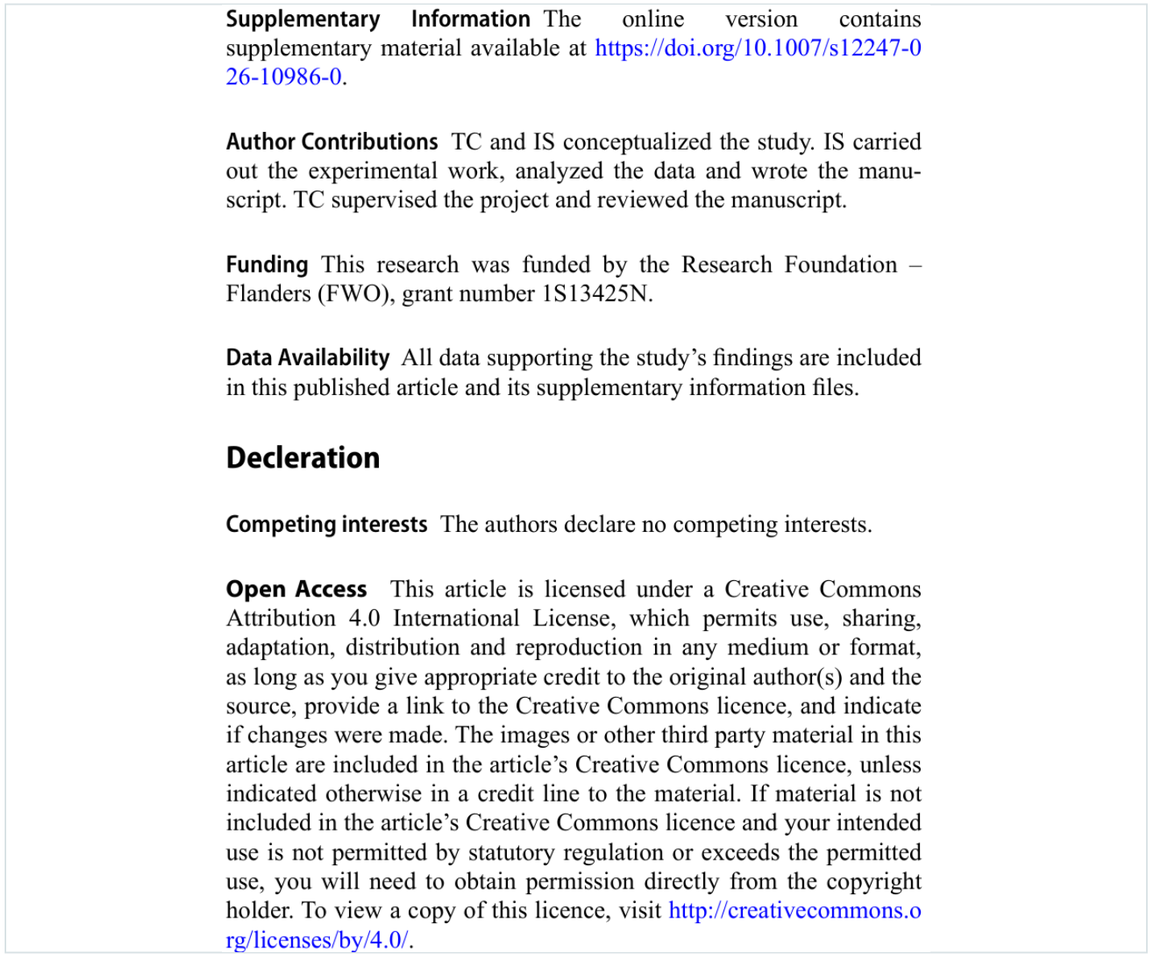
本课件的待验证设备假设

可研究动态基线处理是否改善慢漂移下的判定，同时预设误报代价；也可比较固定阈值与时间序列判定是否改变弱信号漏检。

这些是未来假设，须由独立、含空白与失败样品的数据检验；不能依据本论文直接调低阈值或宣称优于光学方法。

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来源、许可与研究透明度



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Author Contributions TC and IS conceptualized the study. IS carried out the experimental work, analyzed the data and wrote the manuscript. TC supervised the project and reviewed the manuscript.

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Data Availability All data supporting the study's findings are included in this published article and its supplementary information files.

Decleration

Competing interests The authors declare no competing interests.

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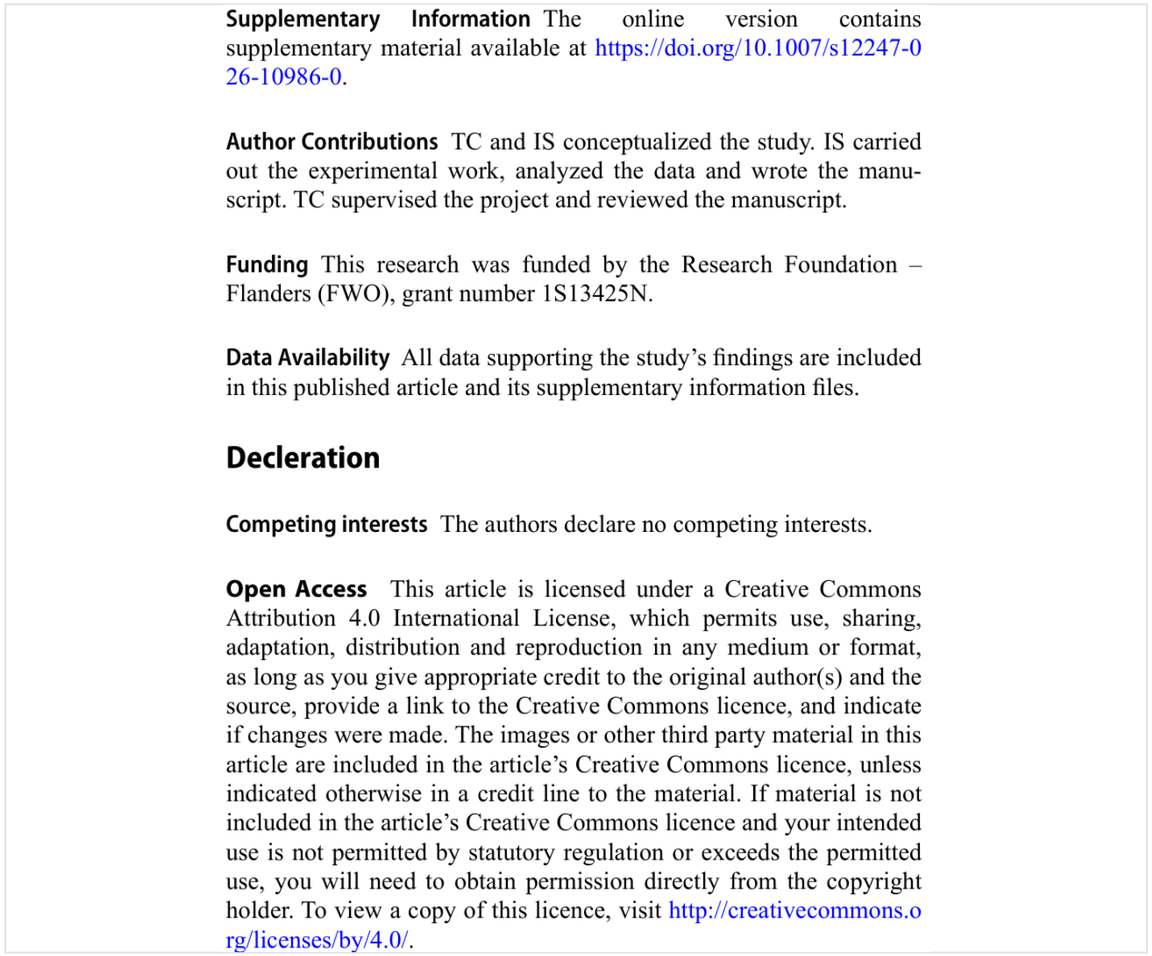
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Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s12247-026-10986-0.

Author Contributions TC and IS conceptualized the study. IS carried out the experimental work, analyzed the data and wrote the manuscript. TC supervised the project and reviewed the manuscript.

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Data Availability All data supporting the study's findings are included in this published article and its supplementary information files.

Decleration

Competing interests The authors declare no competing interests.

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热流阈值、参考容器与数据处理来自本文。热电堆/热沉的通用背景参考厂商公开会议资料；教学方程并非本文披露的专有硬件模型。

制作方法来源

实际阅读no-ai-slop与Scientific Writing 2.3及证据、图表规范。Kassis等：Scientific Agent Skills (2026)，doi:10.48550/arXiv.2609.00065；arXiv v2。

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要点

- 测量代谢热
- 阈值定义报阳
- 保留未检出与分母
- 短报阳不等于早放行

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